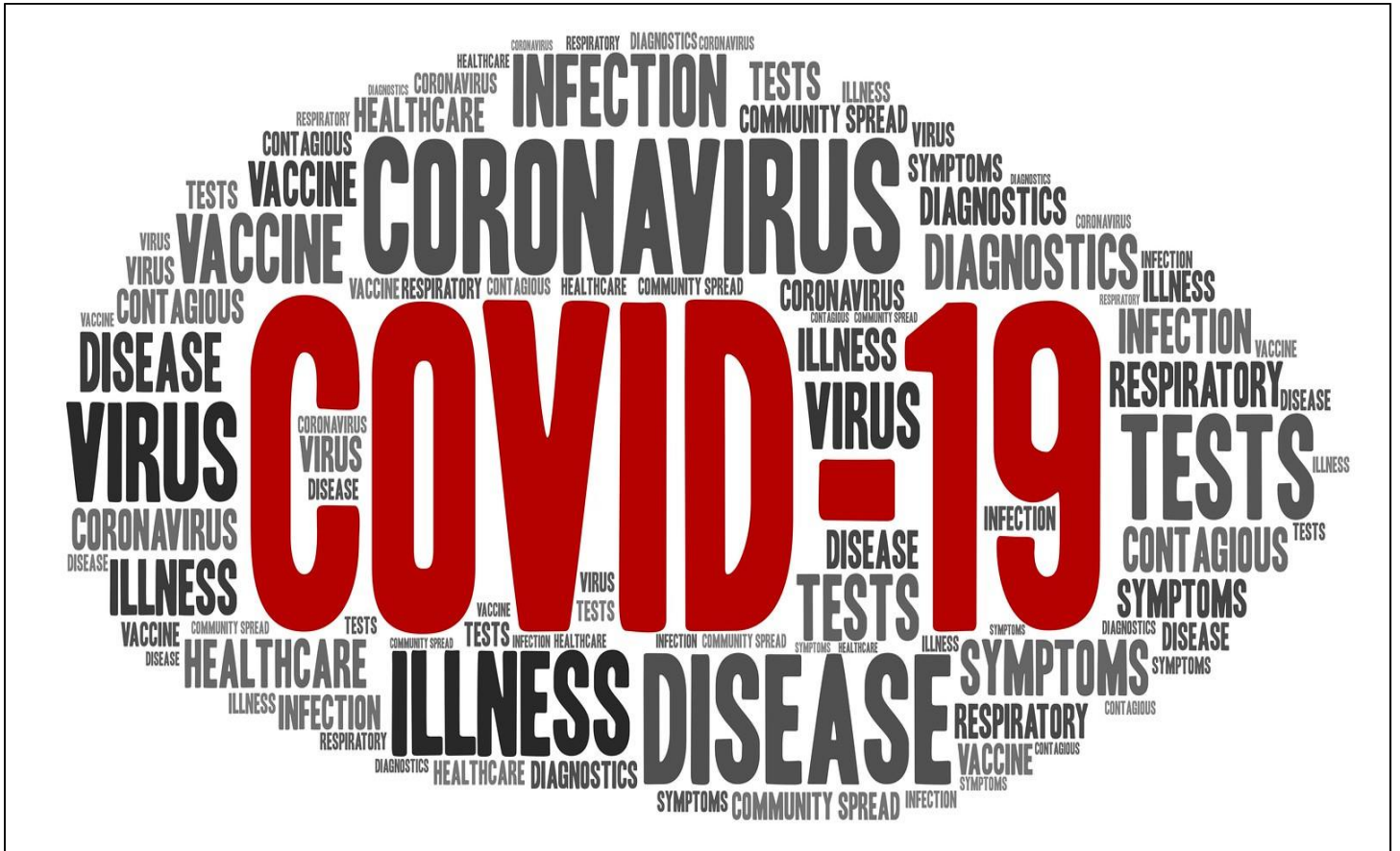


## Covid Data Analysis



Project Group 9

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# 1. Business Understanding

The aim of the report is to analyze the socioeconomic and demographic characteristics of U.S. counties and determine how these factors affected how they responded to the COVID-19 pandemic. To find trends in the pandemic's effects and reactions, we want to use cluster analysis to put counties with comparable profiles together. This analysis's main objectives are to examine how counties' population demographics and economic characteristics are comparable, to uncover significant disparities between the resulting groups, and to pinpoint clusters that were disproportionately affected by COVID-19.

We want to create clusters that represent different degrees of impact by concentrating on age and racial demographics (e.g., the percentages of Black and Hispanic people), economic vulnerability factors (e.g., poverty rates, Gini index, and median income), and pandemic-specific metrics (e.g., death rates and case percentages). To identify whether age or racial groups were more vulnerable, for example, clustering incorporates factors pertaining to age ranges and racial demographics. To comprehend how economic conditions might have influenced county responses and outcomes, economic variables like the unemployment rate and income levels are also examined. To validate the cluster patterns, the hierarchical and k-means clustering methods used in this research offer a reliable comparison between various clustering approaches. We hope to evaluate clustering stability and uncover unique insights on COVID-19 responses at the county level by investigating a variety of feature subsets. This analysis, which adheres to the CRISP-DM framework, highlights socioeconomic vulnerabilities made worse by the pandemic and offers insights that may direct public health actions.

## 2. Data Preparation: [OBJ]

In this analysis, counties from New York, New Jersey, Oregon, and California are used as primary clustering objects. New York County (NY), Essex County (NJ), Malheur County (OR), and Kings County (CA) are among the counties that were chosen. These counties were included because they offer a representative sample of a range of socioeconomic and demographic origins, urbanization levels, and COVID-19 pandemic responses. By grouping these counties together, we hope to spot trends based on common traits, capturing variations in COVID-19 impact and susceptibility across various US areas.

### States and Counties Selected

To ensure a broad range of demographic, economic, and geographic profiles and to improve the depth of clustering analysis, the states and counties that were part of this investigation were chosen. Comparisons that show how these factors may affect COVID-19 outcomes are made possible by the areas that were selected because they represent different degrees of racial and cultural diversity, economic inequality, and urbanization. The main justifications for this choice are:

**Diversity in Socioeconomic Profiles and Demographics:** The chosen counties reflect racial and ethnic diversity, especially in urban areas with significant Black and Hispanic populations, such as Kings County and New York County. The clustering technique is able to pinpoint demographic susceptibility to COVID-19 thanks to this variability.

**Economic Variability:** By including counties from states with varying economic profiles, the analysis looks at how variables like unemployment, poverty rates, and income levels affect the impact of COVID-19. For example, Oregon offers ideas from a more rural economy, whereas California and New York are states with diverse economies.

**Variation in COVID-19 Impact:** The public health responses and pandemic dynamics varied among the states that were chosen. Early on, New York and New Jersey suffered greatly, but California's size led to a range of regional results. Oregon's inclusion permits more variation in pandemic reactions.

**Representation of Both Urban and Rural Areas:** Counties such as Malheur County (rural) and New York County (urban) provide a comparison of how urban and rural areas responded to COVID-19. This choice

makes it possible to analyze variables that could affect pandemic results, like population density and access to medical resources.

**Features Used for Clustering:** Demographic, economic, and health outcome variables make up the clustering features. Age, race, socioeconomic inequality, and the severity of the pandemic in each county are among the COVID-19 risk factors that are captured by these variables. By concentrating on these characteristics, the analysis seeks to classify counties with comparable characteristics and spot trends regarding COVID-19 impact and vulnerability.

| Feature Name             | Scale of Measurement | Description                                            |
|--------------------------|----------------------|--------------------------------------------------------|
| Hispanic Population %    | Ratio                | Percentage of Hispanic population                      |
| Black Population %       | Ratio                | Percentage of Black population                         |
| Age 45-54 Black %        | Ratio                | Percentage of Black population aged 45-54              |
| Age 55-64 Black %        | Ratio                | Percentage of Black population aged 55-64              |
| Age 45-54 Hispanic %     | Ratio                | Percentage of Hispanic population aged 45-54           |
| Age 55-64 Hispanic %     | Ratio                | Percentage of Hispanic population aged 55-64           |
| Poverty Percentage       | Ratio                | Percentage of population living below poverty line     |
| Gini Index               | Interval             | Income inequality index                                |
| Unemployment Rate        | Ratio                | Percentage of population unemployed                    |
| Income < \$10,000        | Ratio                | Percentage of population with income < \$10,000        |
| Income \$10,000-\$14,999 | Ratio                | Percentage of population with income \$10,000-\$14,999 |
| Death Percentage         | Ratio                | Percentage of deaths relative to total population      |
| Median Income            | Ratio                | Median income of the population                        |

Except for the Gini index, which is interval-scaled, these attributes are primarily ratio-scaled. While the interval scale used for the Gini index reflects relative income inequality without a genuine zero point, making it appropriate for evaluating differences across counties, ratio scales enable meaningful comparisons of proportions, such as poverty and death percentages.

A statistical summary of each feature, including the mean, median, standard deviation, variance,

minimum, maximum, and range, is given in the table below. This synopsis provides information on the variability and distribution of the data, which directs the clustering analysis and results interpretation.

| Feature Name               | Mean     | Median  | Std Dev | Variance    | Min   | Max    | Range  |
|----------------------------|----------|---------|---------|-------------|-------|--------|--------|
| Hispanic Population %      | 9.12     | 3.98    | 13.71   | 188         | 0     | 99.18  | 99.18  |
| Black Population %         | 8.9      | 2.13    | 14.44   | 208.64      | 0     | 86.92  | 86.92  |
| Age 45-54 Black %          | 0.59     | 0.18    | 0.92    | 0.84        | 0     | 5.75   | 5.75   |
| Age 55-64 Black %          | 0.53     | 0.13    | 0.88    | 0.78        | 0     | 6.25   | 6.25   |
| Age 45-54 Hispanic %       | 0.51     | 0.21    | 0.83    | 0.69        | 0     | 6.66   | 6.66   |
| Age 55-64 Hispanic %       | 0.35     | 0.12    | 0.68    | 0.47        | 0     | 9.03   | 9.03   |
| Poverty Percentage         | 15.37    | 14.61   | 6.22    | 38.67       | 2.42  | 51.37  | 48.94  |
| Gini Index                 | 0.44     | 0.44    | 0.04    | 0           | 0.33  | 0.6    | 0.27   |
| Unemployment Rate          | 6.33     | 6.03    | 3.04    | 9.27        | 0     | 28.73  | 28.73  |
| Income < \$10,000 %        | 2527.77  | 763.5   | 7869.6  | 61930610.3  | 0     | 201863 | 201863 |
| Income \$10,000-\$14,999 % | 1835.81  | 625     | 5567.39 | 30995866.9  | 0     | 178737 | 178737 |
| Death Percentage           | 0.13     | 0.12    | 0.09    | 0.01        | 0     | 0.84   | 0.84   |
| Median Income              | 49754.16 | 48065.5 | 13154.4 | 173038893.7 | 19264 | 129588 | 110324 |

### 3. Scale of Measurement of the Features:

In this analysis, clustering is generally done using numerical and ratio-scaled characteristics. Meaningful ratio calculations are made possible by ratio-scale variables, which include the percentages of the Hispanic and Black populations, poverty, unemployment, and death rates, as well as income levels. These variables have a true zero point. Ratio-scaled data is appropriate for distance-based measurements in clustering because, for instance, a county with a 20% poverty rate has twice the poverty level of one with a 10% rate. This is because discrepancies between values are similar and interpretable.

One exception is the interval-scaled Gini Index. Because it doesn't have a real zero, this measure only measures income disparity within a range, which makes it perfect for analyzing relative variations in income distribution between counties as opposed to absolute comparisons.

#### Measures for Similarity/Distance

A few distance measures were used to account for the data structure and improve the clustering analysis

because the data was both ratio- and interval-scaled:

**Euclidean Distance:** This measure of the straight-line distance between two points in a multi-dimensional space is frequently applied to continuous, ratio-scaled data. Based on the absolute differences in continuous variables like population and economic percentages, Euclidean distance in this research efficiently represents the general similarity between counties. Its computation entails squaring the differences between every pair of points, making it sensitive to greater distances and highlighting counties with more notable disparities in particular traits.

**Manhattan Distance:** sometimes known as "taxicab" distance, is a method of calculating the sum of absolute differences across each feature dimension, whereas Euclidean distance measures distance in straight lines. This measure offers an alternative viewpoint, especially when dealing with outlier cases. Because Manhattan distance is based on absolute values rather than squared differences, it is less impacted by extreme values. Manhattan distance, while not the main metric, can be helpful for robustness and enhances Euclidean distance by emphasizing linear differences between counties.

**Gower's Distance:** Gower's Distance was chosen as the main distance measure for clustering because of the heterogeneous character of the dataset. Since Gower's Distance normalizes each feature to a [0, 1] scale, it is perfect for datasets that contain both interval and ratio-scaled features. Features like poverty % and income inequality (interval scale) can seamlessly integrate with strictly ratio-scaled features thanks to this normalization, which guarantees that all variables contribute equally to the similarity computation. Gower's Distance allows for a balanced approach in this research that takes into account both minor and significant differences in demographic and economic characteristics. As a result, it is ideally suited to detecting subtle trends in the socioeconomic profiles and COVID-19 impact across counties.

The analysis can take into consideration the mixed data types and accomplish a thorough clustering strategy by utilizing Gower's Distance in conjunction with the Euclidean and Manhattan distances. Every distance meter offers different perspectives:

The degree of variation in continuous variables affects the Euclidean Distance. All feature types are integrated using Gower's Distance, which successfully captures interval- and ratio-scale properties.

By using a combination methodology, the research can find patterns that represent demographic variety, economic situations, and COVID-19 susceptibility, as well as similarities and variations between counties.

## 4. Modeling:

In this analysis, we evaluate patterns within four main feature subsets using k-means and hierarchical clustering methods. Demographic effect, economic disparity, income vulnerability, and a comprehensive combination of demographic and economic factors are all represented by a different subgroup. By identifying groupings of counties with comparable features, these clustering models allow us to get insight into the relationships between various factors and COVID-19 outcomes.

### A. Cluster One: Demographic and COVID-19 Impact (Hispanic Population Percentage, Black Population Percentage, Death Percentage, Age Group Percentages):

**K-Means:** COVID-19 death rates among New York State counties strongly correlate high rates with counties that have a high Hispanic and Black population and particularly counties that are highly urban. K-means clustering here involves the removal of outliers such as Bronx, Queens, Kings, and Richmond counties (Fig1.1) because these jurisdictions demonstrate very high death rates due to reasons more associated with high population density, socio-economic disparities, and increased vulnerability related to being essential workers

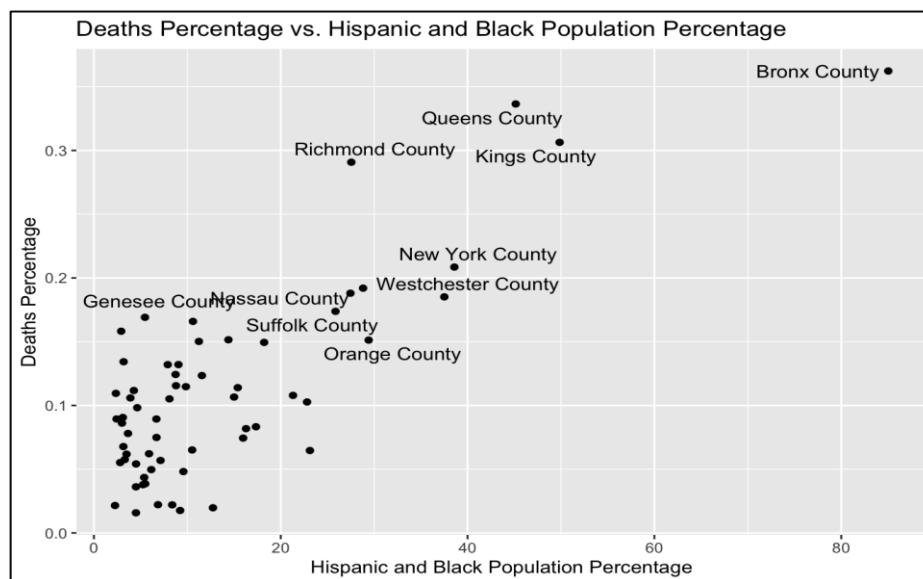


Fig. 1.0



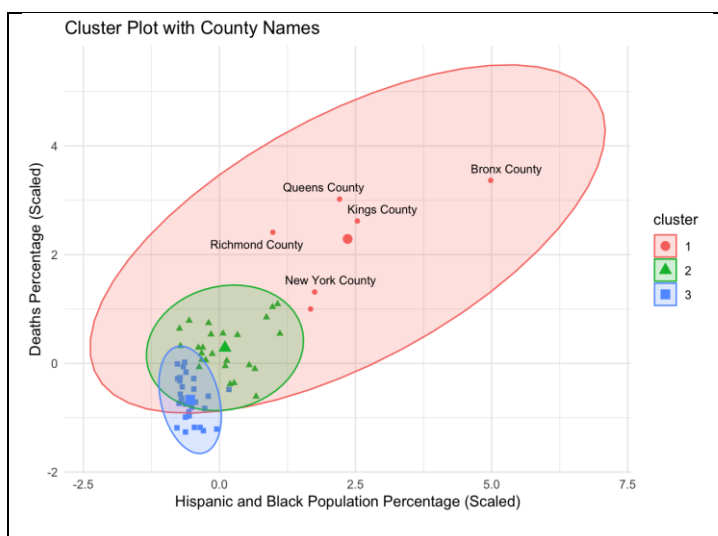


Fig 1.1

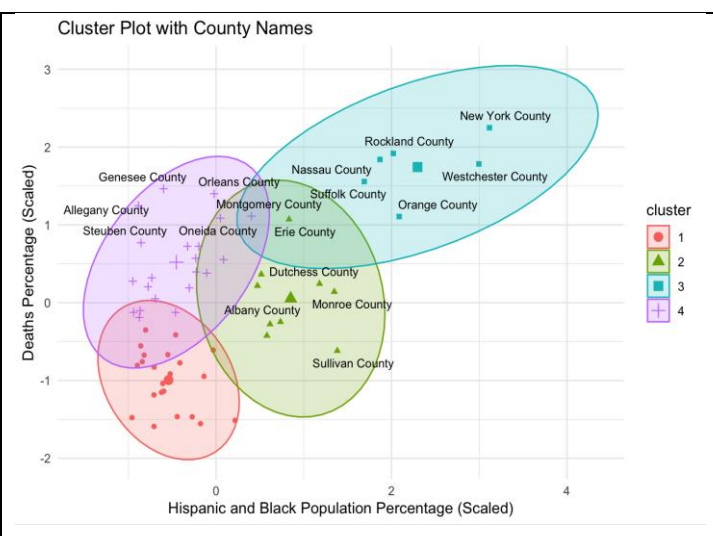


Fig 1.2

After removing the outliers, four clusters with diversified characteristics with the remaining counties were clustered (Fig1.2). Rural counties with lower Hispanic and Black population percentages and low death rates benefited from low transmission rates due to natural social distancing. Clusters of moderate minority populations but with varying death rates point to differences in healthcare access and relative levels of urbanization. This can be an example of that: New York and Westchester (Cluster 3) having the most deaths from COVID-19, given the higher densities and disparities in economics that impede healthcare access within minority communities. Conversely, counties with high death tolls yet low minority representation, such as Orleans and Genesee (Cluster 4), had other risk factors—for instance, older populations or inadequate healthcare resources. This clustering analysis underlines the fact that both population density and racial demographics are important factors in COVID-19 mortality. High Hispanic and Black populations in counties, especially urban areas, face different challenges that could be mitigated in any future health crisis if specific healthcare support and socioeconomic intervention were used appropriately.

To further understand these trends across a broader set of counties, we extended the analysis using hierarchical clustering. This approach included additional outliers from New York, as well as counties from other states, such as CA, NJ, and OR, that displayed similar patterns.

**Hierarchical clustering:** To find trends in the demographic and socioeconomic characteristics linked to pandemic outcomes, I concentrated on grouping COVID-19-affected U.S. counties. Because of their diverse demographic makeup and unique COVID-19 effects, the designated counties—New York County

(NY), Essex County (NJ), Malheur County (OR), and Kings County (CA)—were picked to enable a thorough examination of patterns in various geographical areas. Since each county has sizable Black and Hispanic populations, it is vital to research differences in COVID-19 outcomes according to socioeconomic and ethnic characteristics. To better understand how particular age and ethnic groups may have been more vulnerable to COVID-19, I focused on Hispanic Population Percentage, Black Population Percentage, and age-related features within these groups (e.g., Ages 45-54 and Ages 55-64 for both Hispanic and Black populations) for the first clustering subset. Death Percentage was also taken into consideration to directly examine health implications. This subset sheds light on how each county's demographic characteristics affect its vulnerability to COVID-19.

Using hierarchical clustering with Ward's method and Gower's Distance (to handle the mixed data scales), I grouped counties based on similarities across these demographic variables, revealing clusters with distinct age and ethnicity profiles. The **dendrogram** effectively separates clusters, such as **Cluster 1** (Kings County, CA) with a high Hispanic population aged 45-64, and **Cluster 2** (Essex County, NJ) with a concentration of Black individuals in the same age range. The **heatmap** visually represents feature values within clusters, with strong color intensities indicating high proportions—such as the red tones for the Black population in Essex County, NJ, and the Hispanic population in Kings County, CA. Additionally, the **bar plot** summarizes mean values by cluster, highlighting demographic differences; for instance, Cluster 1 shows a high mean Hispanic Population Percentage (53.7), emphasizing the need for targeted health interventions in Hispanic-majority counties. These findings suggest that counties with specific age and ethnic group concentrations experience different COVID-19 impacts, which align with observed health disparities during the pandemic. Identifying these patterns helps inform policies and resource allocation for vulnerable communities.

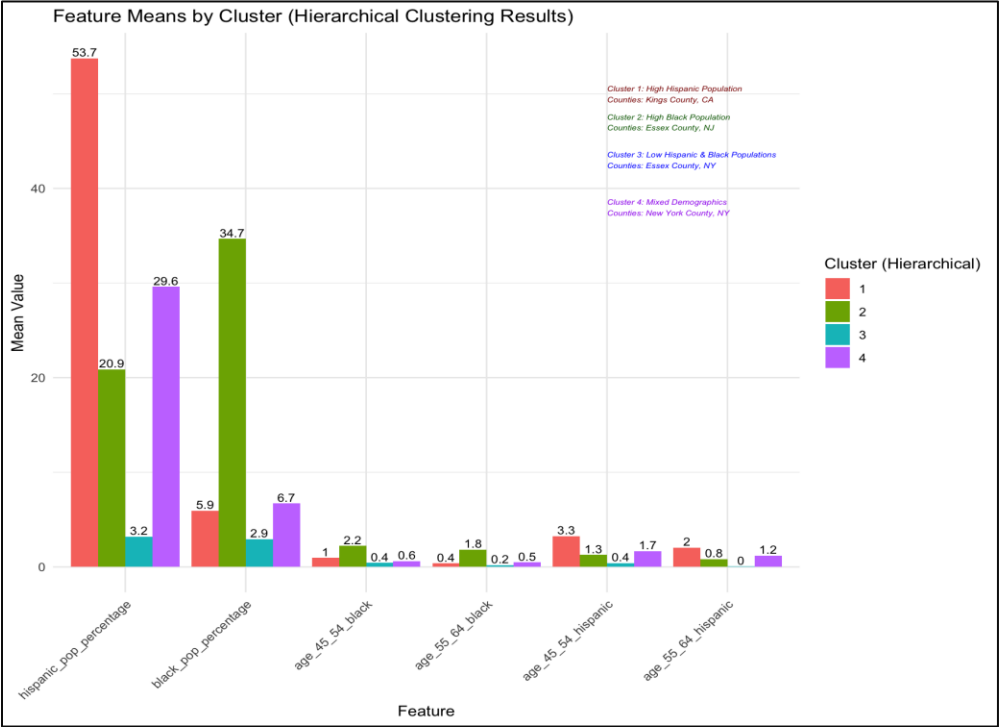
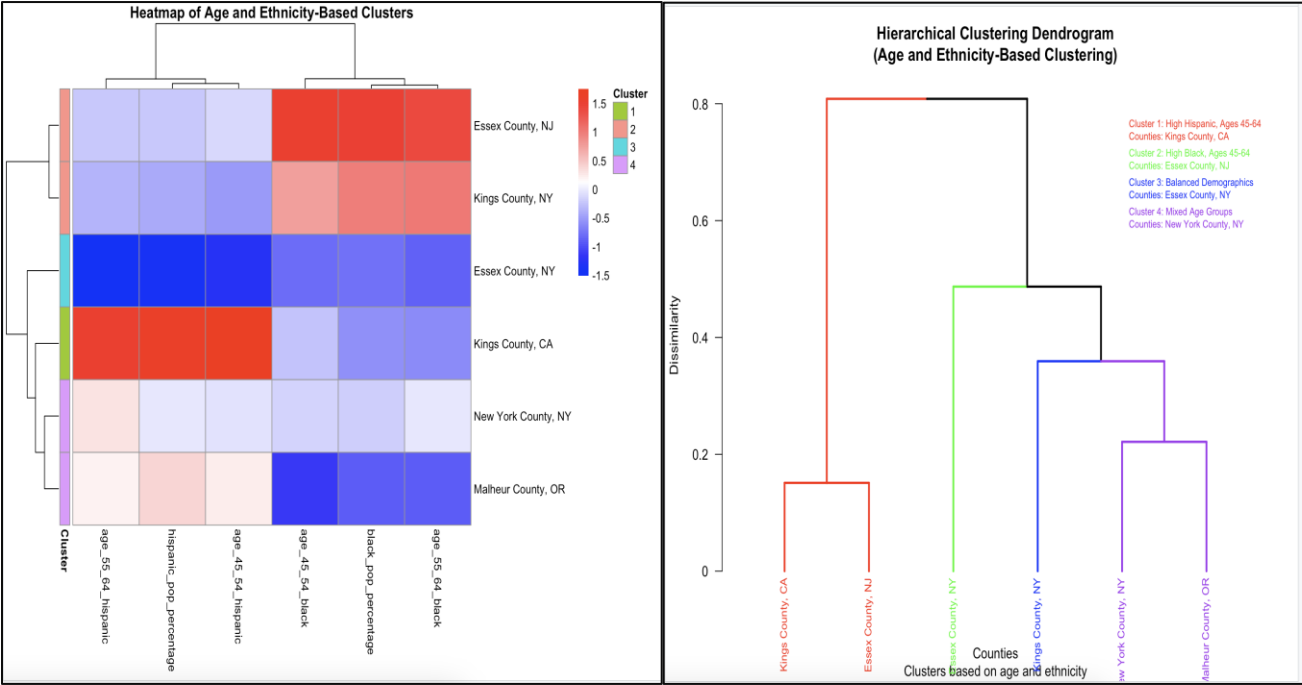
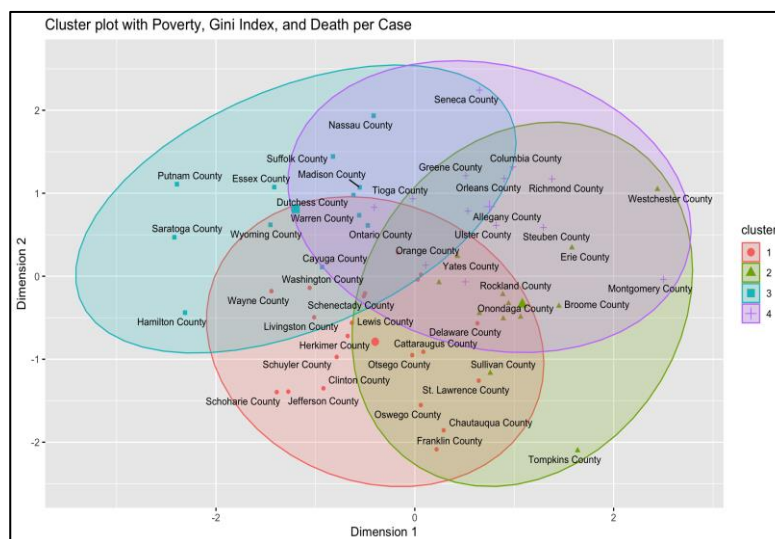


Fig1.3

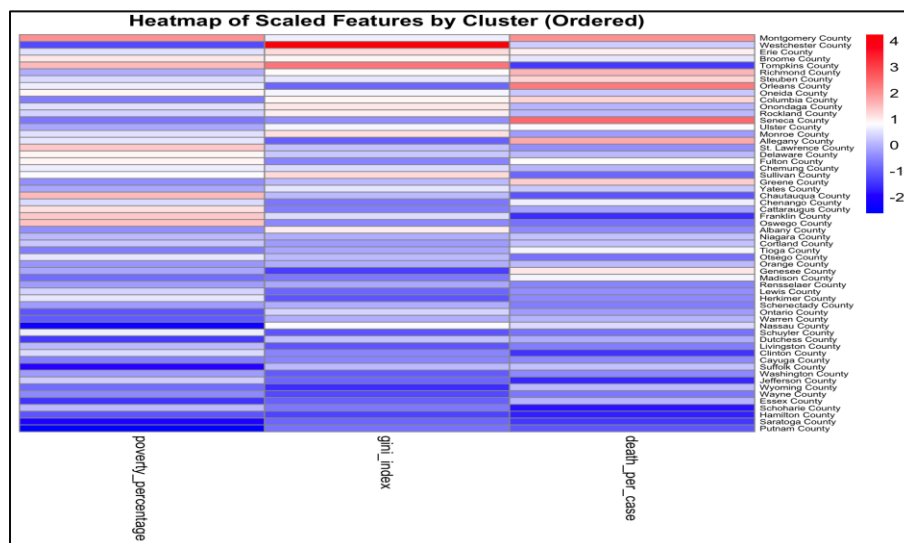
**B. Second Cluster: Economic Vulnerability (Gini Index, Poverty Percentage, and Death Rate per Case):**

We have done K means clustering on all the counties of New York State and then we dive deeper into counties from other states depicting the same behaviors using hierarchical clustering

The cluster plot below classifies New York State counties into four clusters: by poverty percentage, Gini index, and COVID-19 death rate per case. Cluster 1 (Red Circle) consists of counties such as Schuylar, Jefferson, and Schoharie, with relatively high levels of poverty and moderately high COVID-19 death rates. This cluster represents economically vulnerable counties where poor healthcare resources, and a higher poverty level have possibly resulted in inferior health outcomes. Cluster 2 consists of the green triangle of Sullivan, Tompkins, and St. Lawrence counties. All these exhibit moderate levels of both poverty and income inequality but, on the other hand, high COVID-19 death rates. Hence, it is arguable that moderate socioeconomic conditions add to other contributing factors, such as population density or demographics, which together have increased the risk of mortality through COVID-19. Cluster 3, which would be the cluster of the Blue Square, consists of the more affluent counties like Saratoga, Putnam, and Essex. The poverty rates are lower, there is low-income inequality, and the COVID-19 death rates were also low, which suggested that socioeconomic stability could have served as some form of protection against high mortality. Cluster 4 consists of Nassau, Seneca, and Westchester; these have high income inequality, though with differing levels of poverty, and thus have moderate death rates. This may indicate that although income inequality may create some disparity in these counties, other factors such as better healthcare infrastructure or economic resources have countered this effect of income inequality on COVID-19 mortality.



The following heat map gives a better insight into the scaled values of poverty percentage, Gini index, and COVID-19 death rate per case in counties of New York State by clusters. Note that Cluster 1, including counties of Montgomery, Wayne, and Herkimer, exhibits high poverty and a moderate to high death rate, colored red for poverty and death rate, reflecting poorer regions with higher COVID-19 mortality. Counties such as Tompkins, Sullivan, and St. Lawrence have relatively less extreme economic hardships in terms of poverty and income inequality, but higher death rates—a sign of possible other vulnerability factors. Cluster 3, consisting of Putnam, Saratoga, and Essex, demonstrates another color—bluish in appearance—with low rates of poverty and income inequality and low death rates; this reflects the return of health to economic security and equity in income distribution. Cluster 4 is made up of Westchester, Nassau, and Seneca, which have a high Gini index or income inequality, but their poverty levels vary, and the death rates are generally moderate. This perhaps might mean that access to health care or economic resilience limits the effects of COVID-19 on mortality in these counties despite their income disparity. The heat map does a very good job in displaying how combinations of socioeconomic factors influence COVID-19 outcomes across the state.



We used hierarchical clustering to expand the study in order to obtain a more comprehensive view. Based on Economic Factors Clustering focuses on factors like poverty percentage, Gini index (income inequality), and death rate to investigate the relationship between economic vulnerability and COVID-19 results in specific counties. Comparing county profiles and identifying economic patterns influencing COVID-19 impacts are the goals of this grouping.

**Hierarchical Clustering:** Economic Factors-Based Clustering examines the association between economic vulnerability and COVID-19 outcomes in selected counties, with an emphasis on variables such as poverty percentage, Gini index (income inequality), and death rate. The purpose of this clustering is to compare county profiles and find economic trends affecting COVID-19 impacts.

The analysis answers the following important questions: Which counties have comparable economic features? What economic differences exist between the selected clusters? Can we identify economically disadvantaged populations that have been significantly affected by COVID-19? Figure 2.1's dendrogram clusters counties with comparable economic characteristics. For instance, it shows that Cluster 1 (which includes Kings County, CA) has high rates of poverty and mortality, indicating a connection between COVID-19 vulnerability and economic disadvantage. A visual comparison of poverty, income disparity, and death rates among clusters is shown in the heatmap in Figure 2.2. Cluster 1 has greater rates of poverty and death, while Cluster 3 (New York County, NY) has lower rates and moderate inequality. This contrast highlights different economic characteristics and possible cluster-to-cluster differences in COVID-19 impacts. These differences are further shown by the bar plot in Figure 2.3, which confirms that counties with more economic hardship are more severely impacted by COVID-19. Cluster 1 has the greatest poverty rate (20.4%) and death rate (0.4%).

In combination, these visualizations show the relationship between economic factors and the effects of COVID-19. More vulnerable counties are grouped together in Figure 2.1 according to economic similarity. To differentiate between profiles that are economically stable and those that are fragile, Figure 2.2 displays the intensity of each attribute by cluster. The necessity for focused assistance in economically disadvantaged areas is highlighted by Figure 2.3, which confirms that Cluster 1 has the highest rates of poverty and mortality.

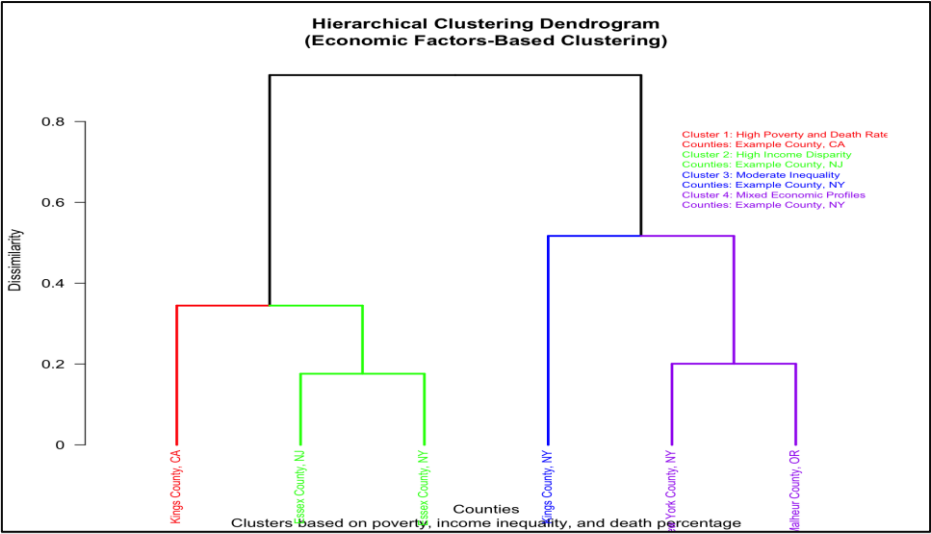


Fig 2.1

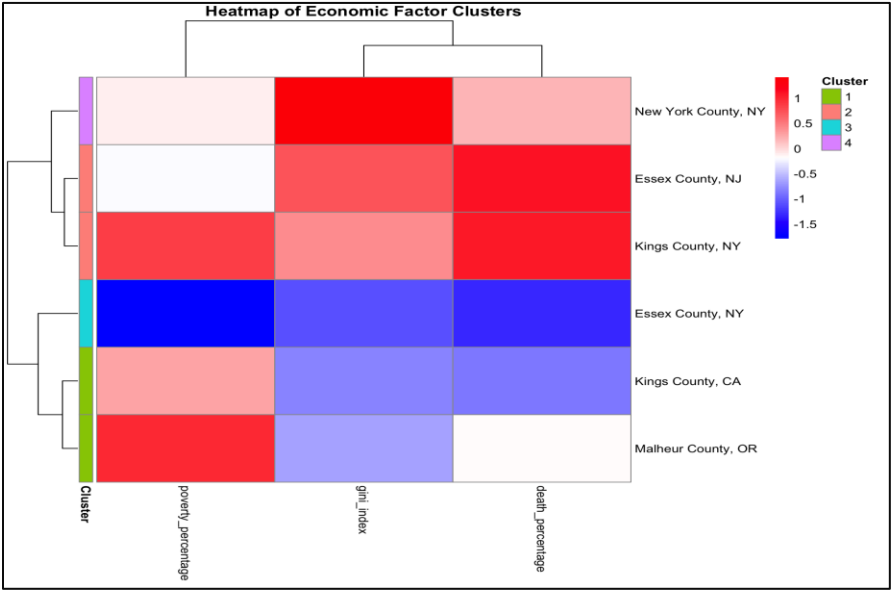


Fig 2.2

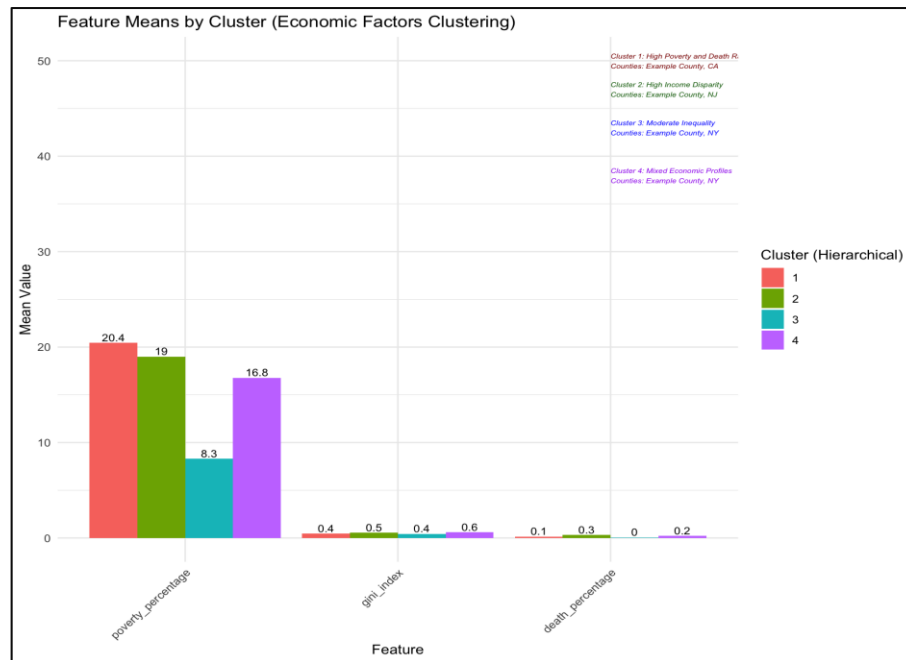


Fig 2.3

### C. Cluster Three: Socioeconomic Hardship and COVID-19 Outcomes (Unemployment Rate, Low-Income Brackets, Death Percentage, Median Income)

To determine a county's vulnerability to COVID-19 effects, we examine it using socioeconomic metrics. The selected characteristics include the following: the unemployment rate, which shows financial hardship and a possible shortage of resources for responding to a pandemic; the income brackets (below \$10,000 and \$10,000–\$14,999), which highlight low-income groups that might have trouble accessing healthcare and other necessities; the death percentage, which calculates the direct toll of COVID-19 on each county.

#### K-Means:

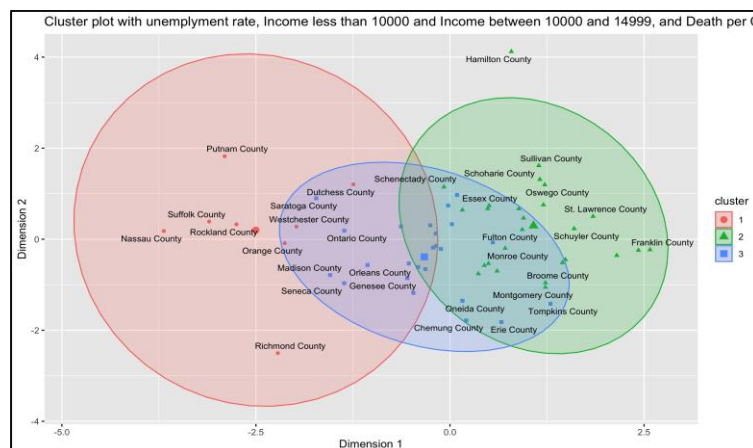
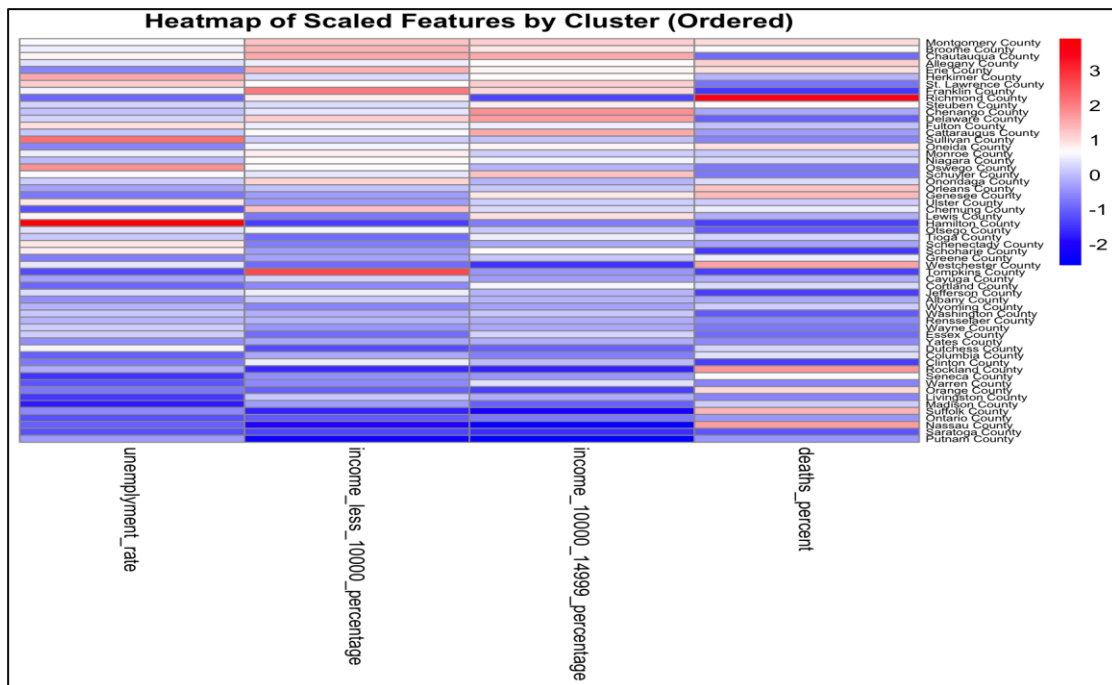




Fig 3.1

The heatmap and cluster plot reveal distinct socioeconomic patterns among New York State counties based on unemployment rate, income distribution, and death rate related to COVID-19. The counties are grouped into three clusters using k-means clustering, showing that some areas have high unemployment and lower income levels, while others, particularly wealthier counties, show a different mix of socioeconomic challenges. Cluster 1 includes wealthier counties like Richmond and Nassau, which display lower low-income percentages but higher unemployment and death rates. Cluster 2 includes a range of counties with moderate income, unemployment, and death rates, while Cluster 3 highlights counties with lower income levels and potentially higher unemployment rates, indicating areas that may be more vulnerable to pandemic-related hardships.

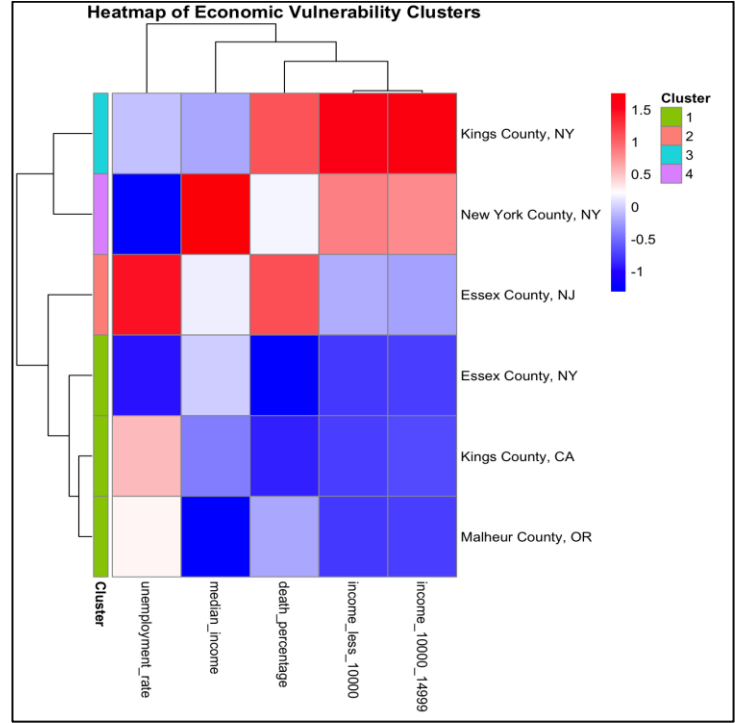
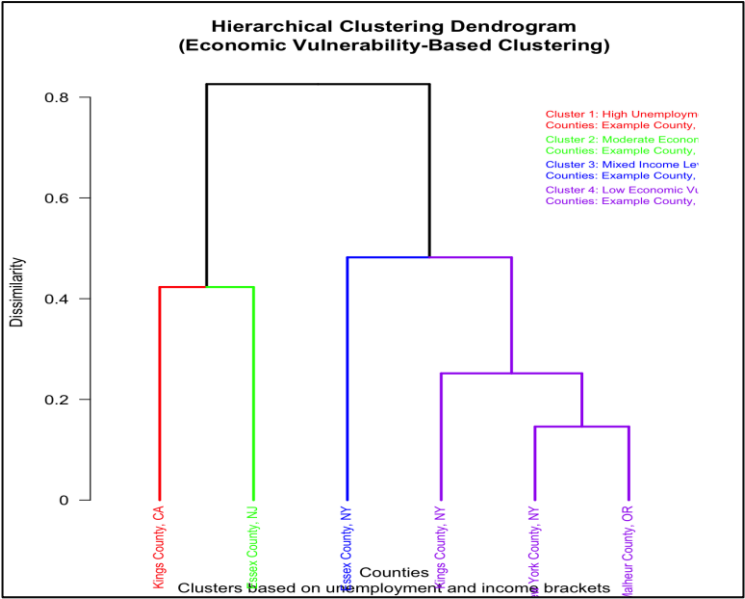
Based on these insights, targeted interventions can be designed to address the distinct needs of each cluster. Cluster 3 counties, with lower income and potentially higher unemployment, may benefit from economic support and increased healthcare access to mitigate vulnerabilities. Cluster 1 counties, though wealthier, may require specialized support to manage higher death rates and unemployment. Meanwhile, Cluster 2 counties could benefit from a balanced allocation of resources, as they display moderate socioeconomic characteristics. This analysis emphasizes the importance of tailored approaches to COVID-19 response efforts, helping New York State allocate resources where they are needed most.

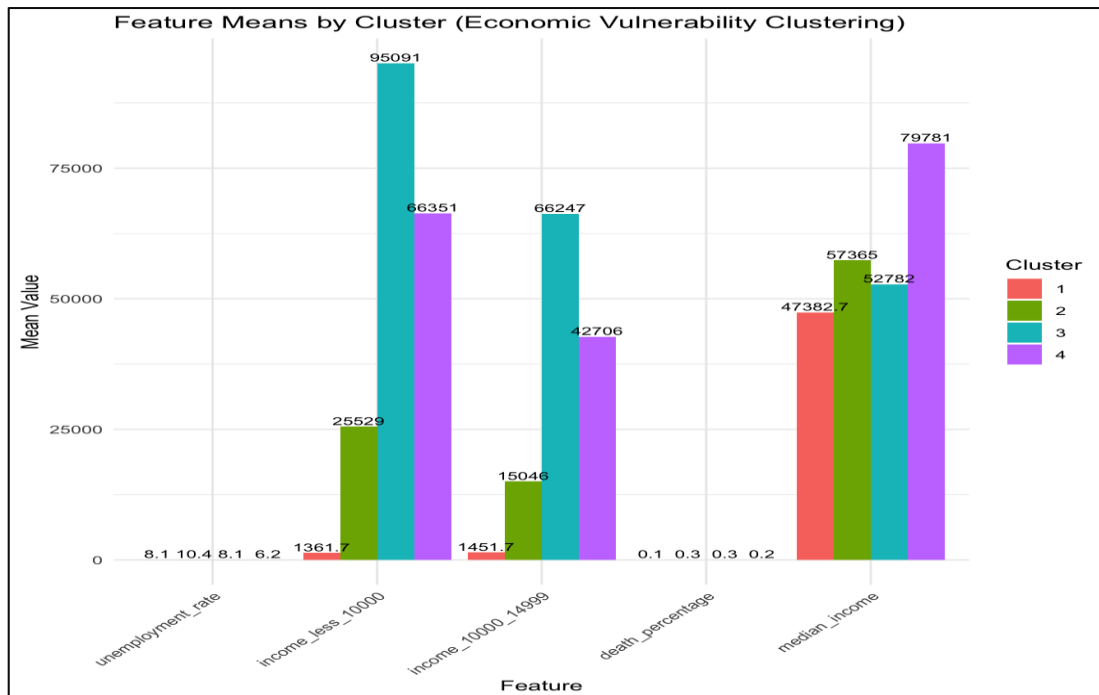


To get a more complete picture of economic vulnerability in connection to COVID-19 outcomes, we extended the study using hierarchical clustering. To examine how economic hardship affects COVID-19 impacts counties, this Economic Vulnerability Clustering looks at variables such as the unemployment rate, income bands (below \$10,000 and \$10,000–\$14,999), death proportion, and median income. By comparing county profiles and spotting socioeconomic trends that might be linked to COVID-19 mortality, this method provides a better understanding of which economically vulnerable groups have been most severely impacted.

### **Hierarchical Clustering**

To determine a county's vulnerability to COVID-19 effects, we examine it using socioeconomic metrics. The selected characteristics include the following: the unemployment rate, which shows financial hardship and a possible shortage of resources for responding to a pandemic; the income brackets (below \$10,000 and \$10,000–\$14,999), which highlight low-income groups that might have trouble accessing healthcare and other necessities; the death percentage, which calculates the direct toll of COVID-19 on each county; and the median income, which provides a more comprehensive indicator of economic stability. A hierarchical grouping of counties based on these economic vulnerability indicators is shown in the Dendrogram (Figure 3.1), which clearly distinguishes between counties with more stable profiles and those with high levels of economic distress. A visual breakdown of each county's values across the chosen features is shown in the Heatmap (Figure 3.2), which highlights the economic discrepancy across counties and draws attention to those with low incomes and high unemployment rates that may have a harder time coping with the pandemic's effects. A summary of average values across clusters is also provided by the Feature Means by Cluster (Figure 3.3), which highlights the clear separation of counties with lower income and more unemployment from those with less economic vulnerability.





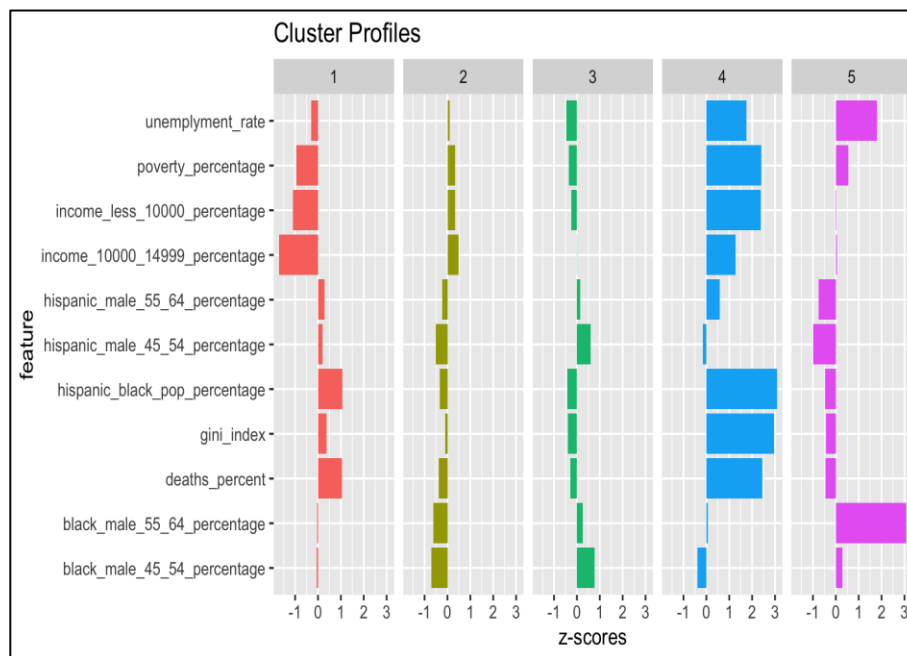
#### **D. Cluster Four: Comprehensive Demographic and Economic Clustering (Hispanic Population Percentage, Black Population Percentage, Poverty Percentage, Gini Index, Unemployment Rate, Income Brackets, Death Percentage)**

The goal of Cluster FOUR, "Comprehensive Demographic and Economic Clustering," is to examine counties by integrating economic and important population demographics in order to comprehend how these attributes affect COVID-19 outcomes. Age distribution within ethnic groups (ages 45-54 and 55-64 for both Black and Hispanic populations), poverty percentage, Gini index (for income inequality), unemployment rate, death percentage, income brackets (percentage of population earning below \$10,000 and between \$10,000-\$14,999), and ethnicity indicators (Hispanic and Black population percentages) are among the many features included in this clustering. To convey total income levels, median income is also included.

#### **K- means clustering**

The plot profiles five clusters of New York counties based on the socioeconomic and demographic features considered, each depicting specific patterns. Cluster 1 depicts slightly higher rates of unemployment, poverty, and COVID-19 deaths, together with high rates of Black and Hispanic males in some of the age groups, indicating moderate socio-economic challenges. Cluster 2 presents high poverty

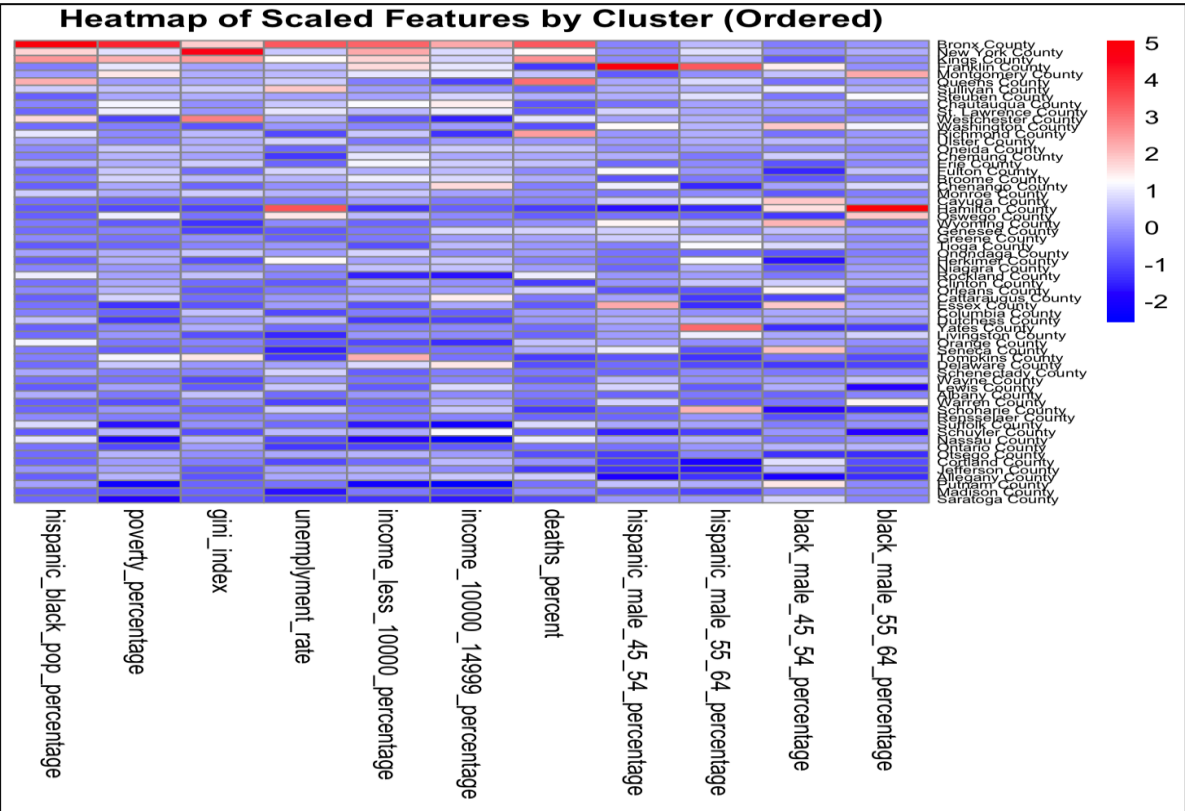
and low-income levels, thus depicting poor areas. Cluster 3 has high income inequality, so it reflects those areas that have marked or heightened socio-economic disparities. Cluster 4 has a high death percentage from COVID-19 and is characterized by high age or ethnic demographics. So, it highlights the especially vulnerable populations. Finally, Cluster 5 has the least variation across all other features, hence reflecting a balanced socioeconomic and demographic characteristic across the counties. These clusters provide a granular insight into county-level vulnerabilities that should be used to inform the targeting of interventions into addressing their unique needs.



The heat map visualizes the scaled values of several socioeconomic and demographic features for New York counties, organized by their assigned cluster. Each feature-poverty rate, unemployment rate, brackets of income, demographic percentages, and COVID-19 death rates-is represented by color, on a scale from blue for the lower value to red for the higher value. Counties are sorted by cluster, enabling us to see shared characteristics and differences within each group.

Most notably, high poverty and income inequality, as indicated by higher values for the Gini index, are found in Bronx County, New York County, and Kings County by their intensified red colors in those columns. These would imply severe economic challenges such as high unemployment rates and an elevated percentage of income under \$10,000, characterizing the economically distressed regions, which are found in Montgomery and Chautauqua Counties. On the other hand, it may be observed that Hamilton County has high z-scores in Hispanic and Black percentage of population; hence, it is highly

distinctive concerning demography. In contrast, counties like Madison and Saratoga have predominantly lower scores consistently across most of the features. For instance, economic and demographic characteristics are relatively more stable. (WRONG INFO FROM NET, NEED EDITING FROM monarch)



**Hierarchical clustering**

The counties are grouped according to these economic and demographic characteristics in the hierarchical clustering dendrogram (Figure 4.1). Counties in Cluster 1 (red) have high rates of poverty and a large Hispanic population, indicating serious economic difficulties. The counties in Cluster 2 (green) have high unemployment rates and a large Black population, indicating that the Black community is economically vulnerable. While Cluster 4 (purple) consists of counties with a diverse population and no discernible economic or ethnic concentration, Cluster 3 (blue) shows counties with a comparatively high median income, suggesting financial stability. A scaled comparison of feature values across each cluster is shown in the parallel coordinates plot (Figure 4.2). Cluster 1 (High Hispanic, High Poverty) has a lower median income and more poverty, while Cluster 3 (High Median Income) has a comparatively higher median income and reduced economic vulnerability. This graphic aids in identifying patterns within each cluster. The socioeconomic differences between clusters are highlighted by this research, which raises

the possibility that counties with greater rates of unemployment, income disparity, or poverty may have experienced more severe economic hardships during COVID-19, especially for some ethnic groups.

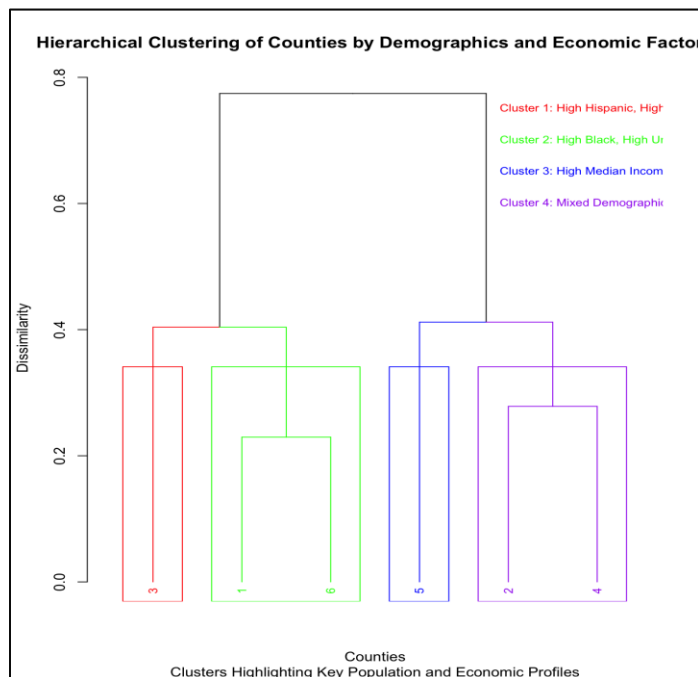


Fig 4.1

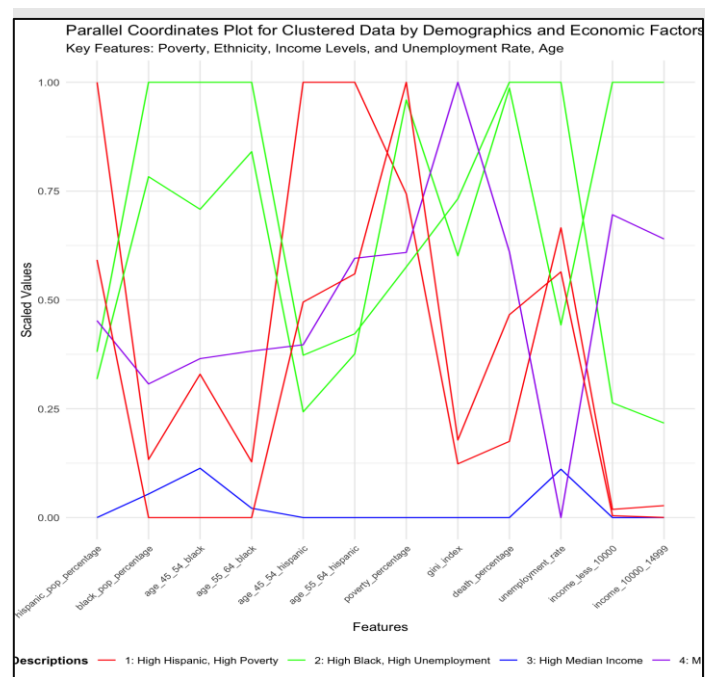


Fig 4.2

## Determining Suitable number of clusters for each method

### A. Cluster One: Demographic and COVID-19 Impact

#### K means

To determine the optimal number of clusters, we applied the Within-Cluster Sum of Squares (WCSS) and Dunn Index methods. The Elbow Method plot of WCSS (Figure 1) indicates an ideal cluster count at  $k = 4$ , where the red line highlights the "elbow" point, suggesting a balance between compactness and cluster distinctiveness. The Dunn Index, however, suggests  $k = 6$  as the optimal number, as it yields better-separated clusters with higher inter-cluster distance relative to intra-cluster compactness. To refine our decision, we examined cluster distribution percentages for  $k = 4$ , we observed that 36.21%, 10.34%, 15.52%, and 37.93% of counties fall into clusters 1 through 4, respectively. This balanced distribution at  $k = 4$  provides meaningful group sizes, making it an effective choice for classifying counties based on racial population distribution and COVID-19 death percentages. Thus, based on the WCSS analysis, balanced distribution, and interpretability of clusters, we determined that  $k = 4$  is the most appropriate cluster count for this analysis.

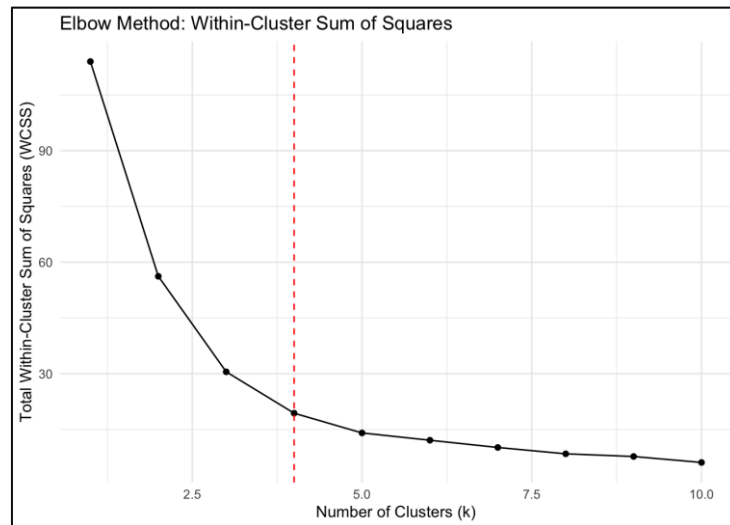


Fig 1. WCSS

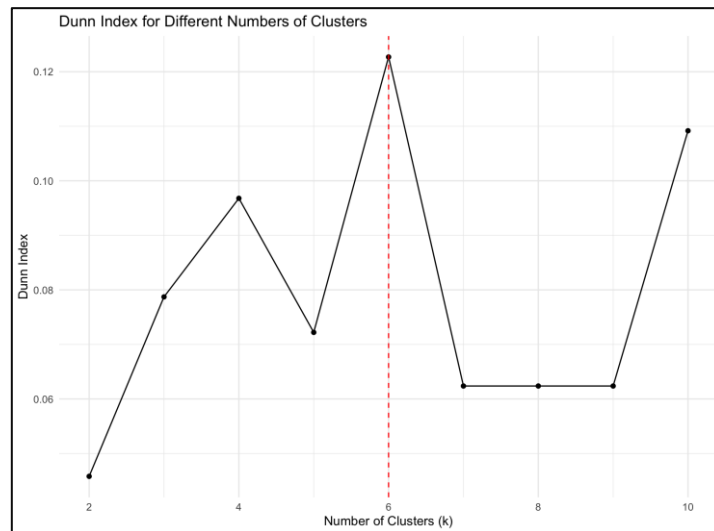


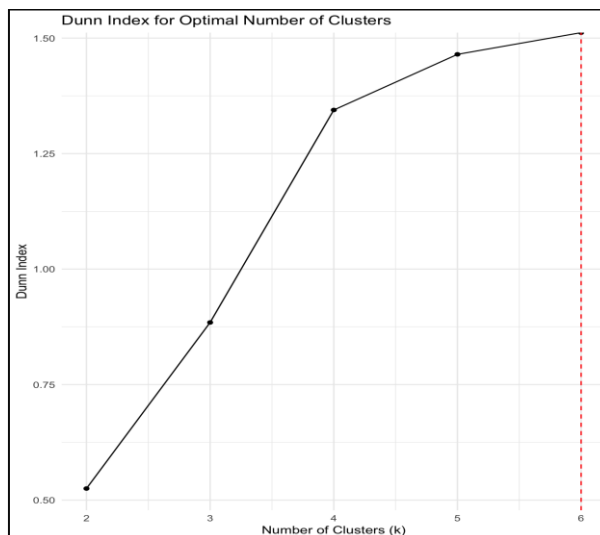
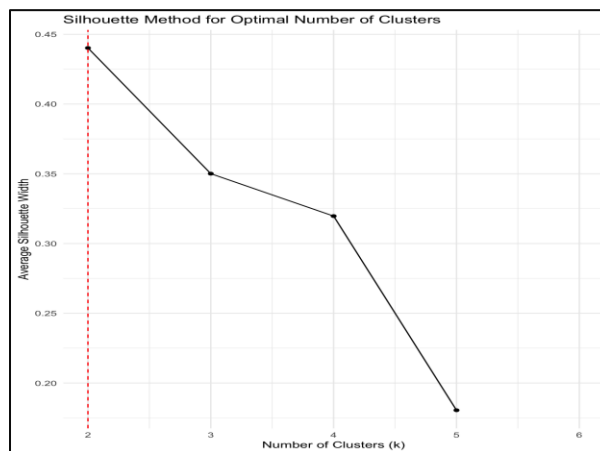
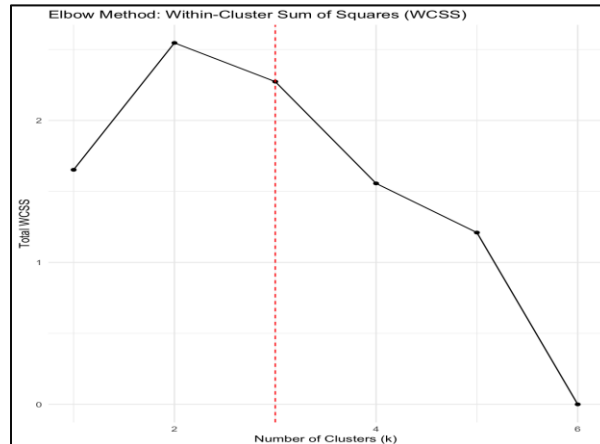
Fig 1. Dunn Index

### Hierarchical Clustering:

We concluded that the optimal number of clusters for this research was  $k = 4$ , based on evaluation techniques for multiple clustering. While the Silhouette Method suggests  $k = 2$  for the most coherent clustering, the Elbow Method displays a discernible "elbow" at  $k = 3$ , indicating a balance between compactness and interpretability. The ideal cluster separation is indicated by the Dunn Index when  $k = 6$ . On the other hand, using  $k = 4$  yields a balanced distribution among clusters, capturing significant demographic information about age and race compositions. In addition to highlighting distinct groups, the



hierarchical dendrogram at  $k = 4$  enables us to understand clusters based on balanced, mixed, high Black, and high Hispanic demographics. This decision is in line with the balanced cluster sizes and the interpretability of demographic trends, which makes  $k = 4$  a good count for categorizing counties according to age-related and ethnic characteristics in COVID-19 results.



## Cluster Two: Economic Vulnerability

Since Cluster 1, focusing on Hispanic and Black population and death percentages, performed well with cluster count determined by comparing the WCSS and Dunn Index plots, we will now apply the same approach to Cluster 2. Here, we will also explore with two methods: the Within-Cluster Sum of Squares (WCSS) and Dunn Index in order to determine the optimal number of clusters. The Elbow Method plot for WCSS (Figure 2. wcss) indicates that  $k = 4$  is the best cluster count, as the "elbow" point (indicated by a red line) marks a good balance between cluster compactness and distinctiveness. Meanwhile, the Dunn Index suggests  $k = 7$  as the optimal count. To further sharpen our decision, we looked at the distribution percentages of counties. For  $k = 4$ , the distribution is balanced with clusters containing, respectively, 36.21%, 20.69%, 22.41, and 20.69% of the counties. In contrast,  $k = 7$  has a much more unbalanced distribution, with the percentages ranging from 1.72% to 27.59% across clusters. This gives meaningful group sizes with balanced distribution at  $k=4$ , hence is effective in the categorization of the counties based on variables such as income disparity, GINI index and poverty percentage, and COVID-19 death per case. Thus, considering the WCSS analysis, balanced distribution, and interpretability, we have determined that  $k=4$  would be the most appropriate number of clusters for this analysis.

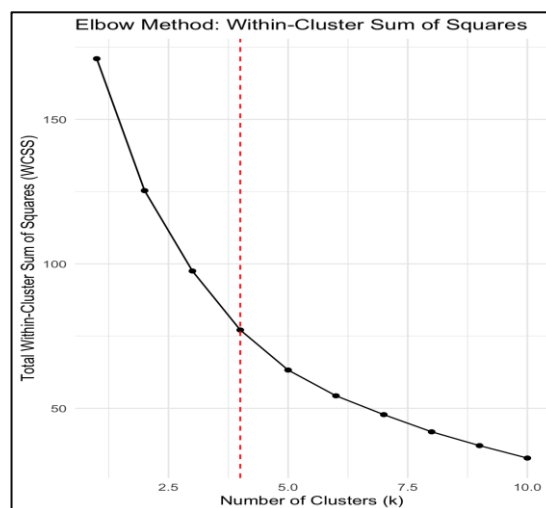


Fig 2.WCSS

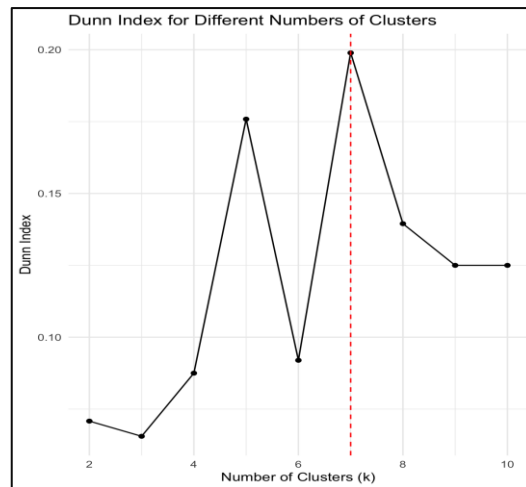
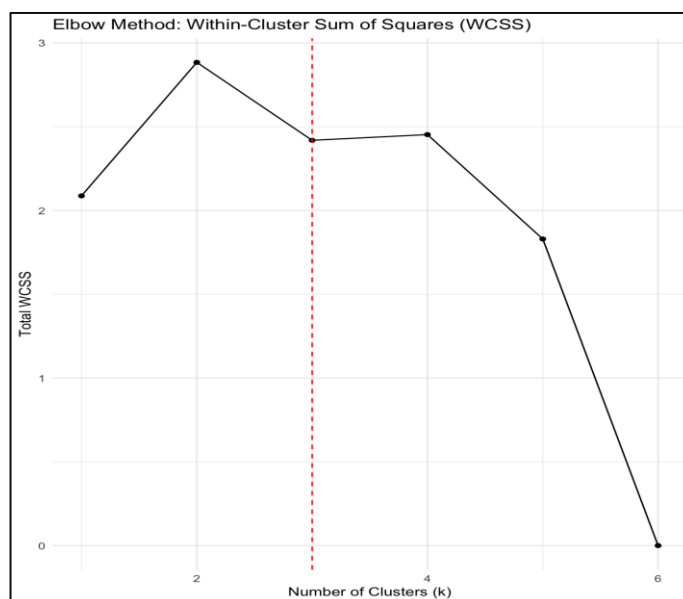
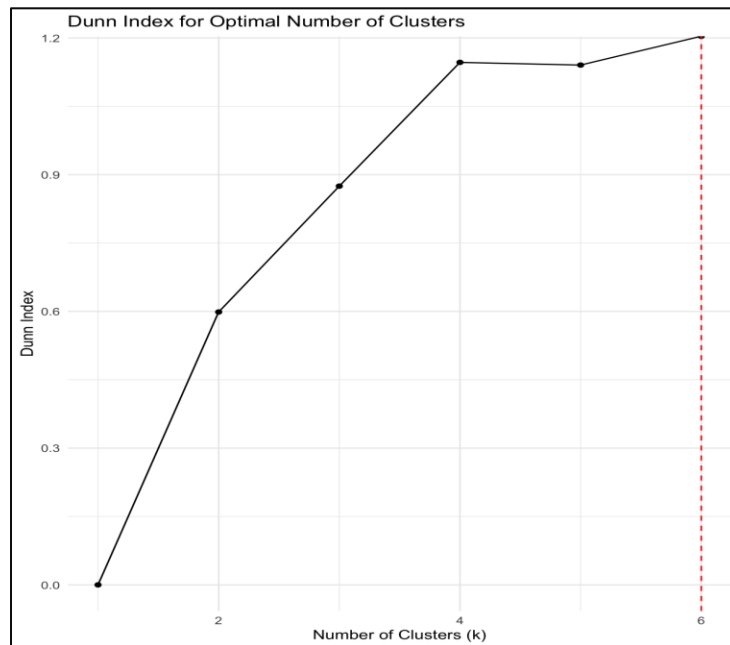
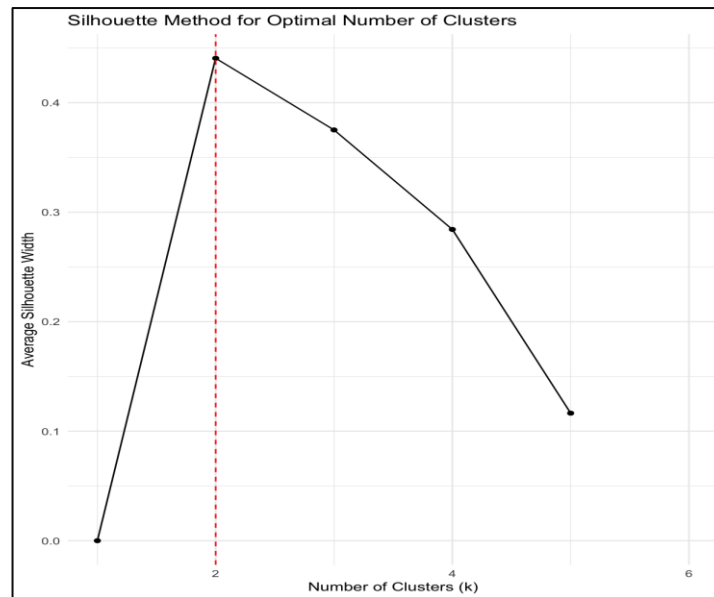


Fig 2. Dunn Index

### Hierarchical clustering:

After analyzing, we evaluated the WCSS, Silhouette, and Dunn Index methods to identify the ideal cluster count. According to the WCSS Elbow Method,  $k = 3$  offers a compact clustering solution that strikes a balance between cluster distinctiveness and compactness. The Silhouette Method indicates a definite separation at  $k = 2$ , although  $k = 3$  is also possible. Although the Dunn Index peaks at  $k = 6$ , its steady rise suggests that  $k = 3$  is a balanced distribution choice that may be understood. In the end,  $k = 3$  provides useful clusters that efficiently classify counties according to COVID-19 death rates, the Gini index, and poverty percentage, giving our analysis a useful profile of economic vulnerability.



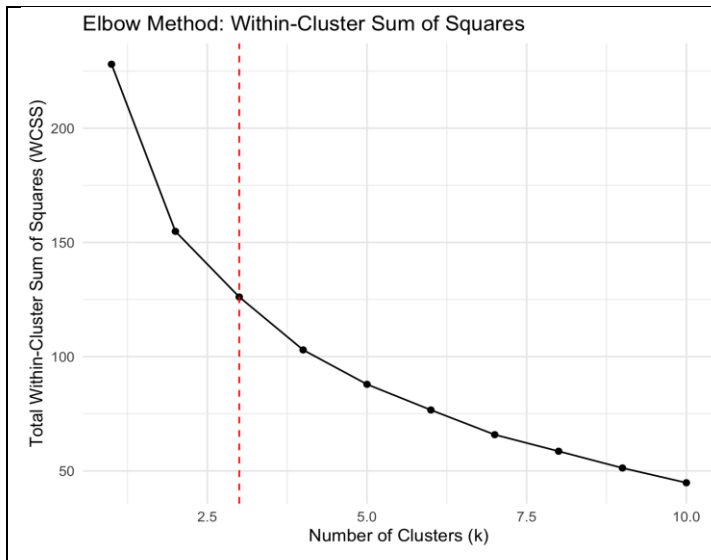


## C. Cluster Three: Socioeconomic Hardship and COVID-19 Outcomes

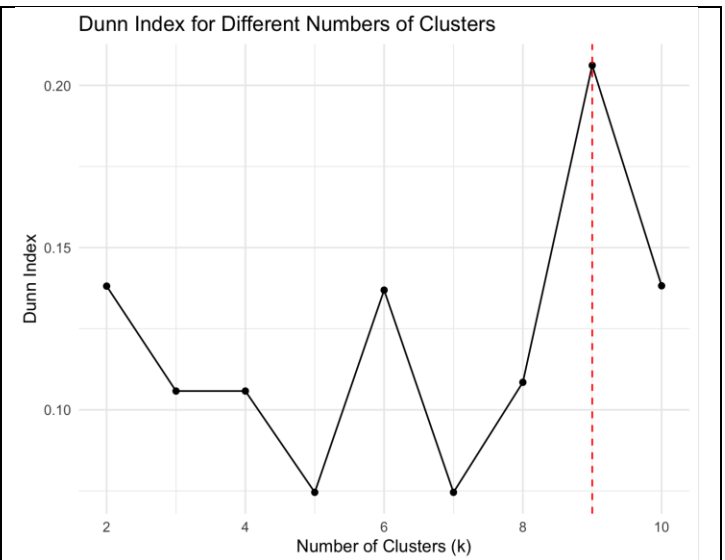
### K means

For cluster 3 I looked at the Within Sum of Squares and Dunn Index to figure out the correct number of clusters. As we can observe from the below WCSS plot, the potential knee is observed at  $k=3$  (where the red line is cutting). This means that the WCSS plot suggests clustering the data with 3 clusters only. Moving on to the Dunn Index Plot, it shows that clustering with nine clusters would be a better option. To solve this Delima, we again calculated the distribution percentages of the counties (data points) over 3 clusters and

9 clusters SIMULTANEOUSLY. For 3 clusters we are getting 44.83% 13.79% and 41.38% data distribution while with 9 clusters we are getting 1.72%, 12.07%, 24.14%, 1.72%, 1.72%, 12.07%, 18.97%, 6.90% and 20.69%. After analyzing the results, we believe that we saw a reasonable clustering of the data with 3 number of clusters.



**Fig 3. Wcss**

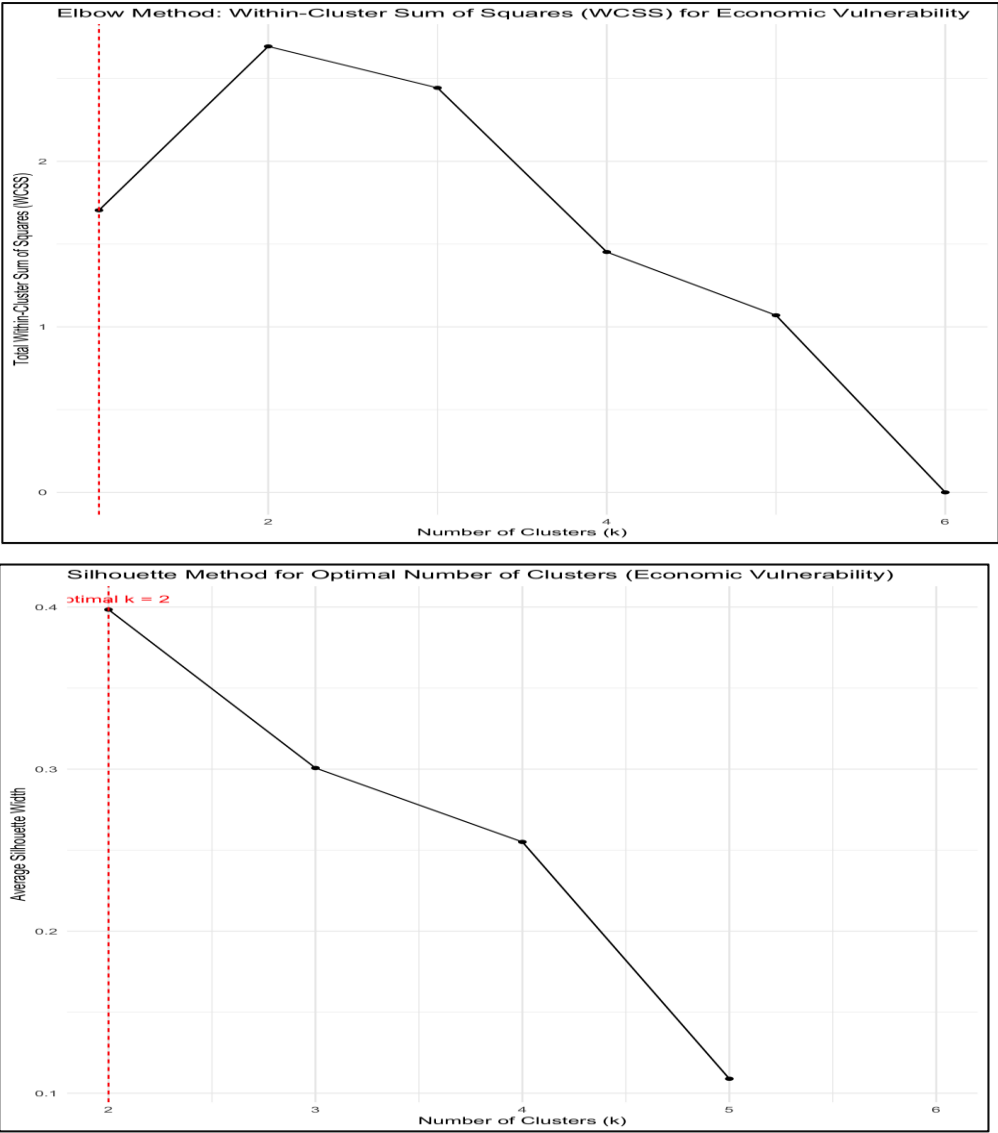


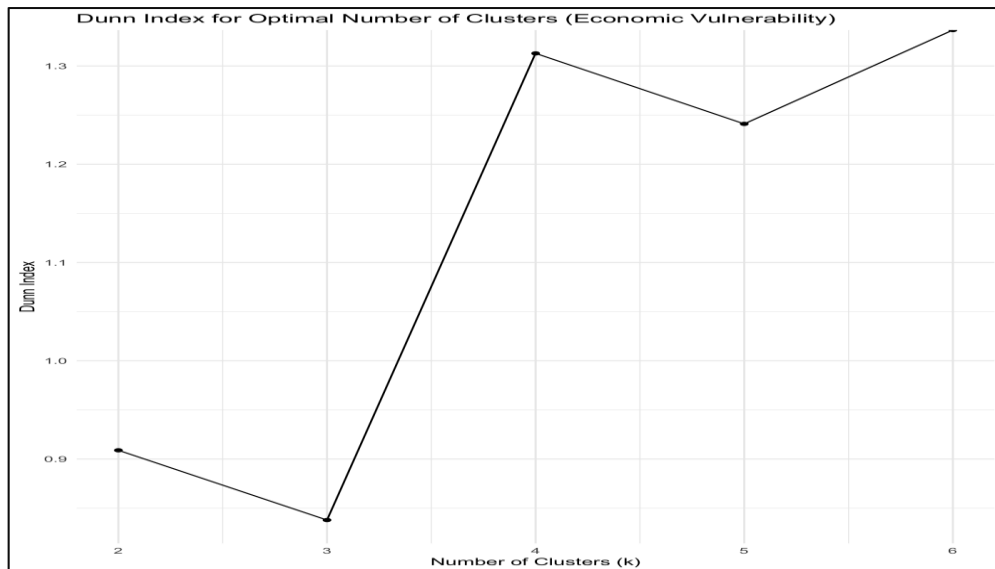
**Fig 3. Dunn Index**

## hierarchical clustering

Economic Vulnerability, we investigated the appropriate number of clusters using three methods: the Elbow Method, Silhouette Width, and Dunn Index, each of which provided a distinct perspective on the clustering structure. Around  $k = 3$ , where a slight curve in the WSS (Within-Cluster Sum of Squares) graph suggested a balance between cluster compactness and separation, the Elbow Method proposed an optimal cluster number. However, based on maximizing the average silhouette width, the Silhouette Width showed an ideal  $k = 2$ , indicating that two clusters offer the most distinct division between groups. The Dunn Index plot, on the other hand, revealed a peak at  $k = 4$ , suggesting that four clusters could more accurately identify discrete groups of economic risk and capture more nuanced variations in the data. We chose  $k = 4$  for this analysis based on the visualization of the hierarchical dendrogram, which indicated a more detailed and understandable clustering structure at this level. Groups with low income and high unemployment, moderate economic vulnerability, mixed income levels, and low susceptibility could not have been distinguished with fewer clusters, but we were able to do so with  $k = 4$ . Compactness (Elbow), cohesiveness (Silhouette), and separation (Dunn Index) were the three data

features that each clustering approach emphasized. By combining these insights, we were able to determine that  $k = 4$  was the most significant and comprehensible option for our economic vulnerability analysis. This choice supports our analytical objectives by enabling us to strike a balance between quantitative measurements and the qualitative framework the dendrogram offers.





## D. Cluster Four: Comprehensive Demographic and Economic Clustering

### K means-

Upon examining the Within-Cluster Sum of Squares (WCSS) plot, it is difficult to identify a clear "elbow" point, making it challenging to determine the optimal number of clusters. The absence of a distinct knee in the plot suggests that the WCSS metric may not provide a definitive answer in this case. Furthermore, while the Dunn Index plot (Figure 4) suggests an optimal cluster count of only 2, this solution is not suitable for our analysis. Having only two clusters does not provide the level of differentiation or granularity required for our objectives.

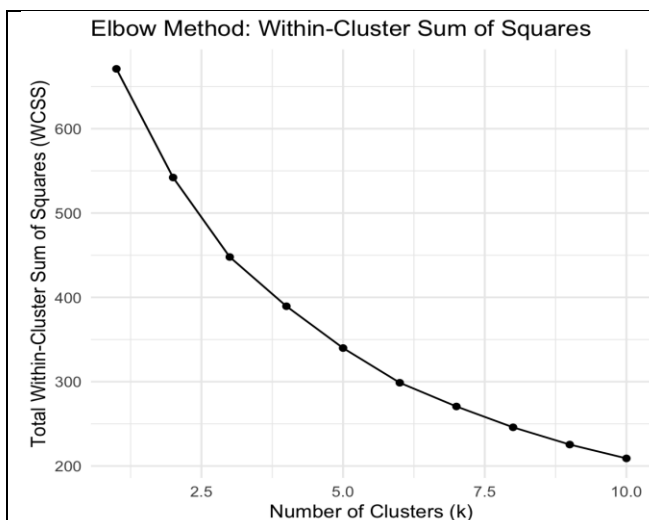


Fig 4.Wcss

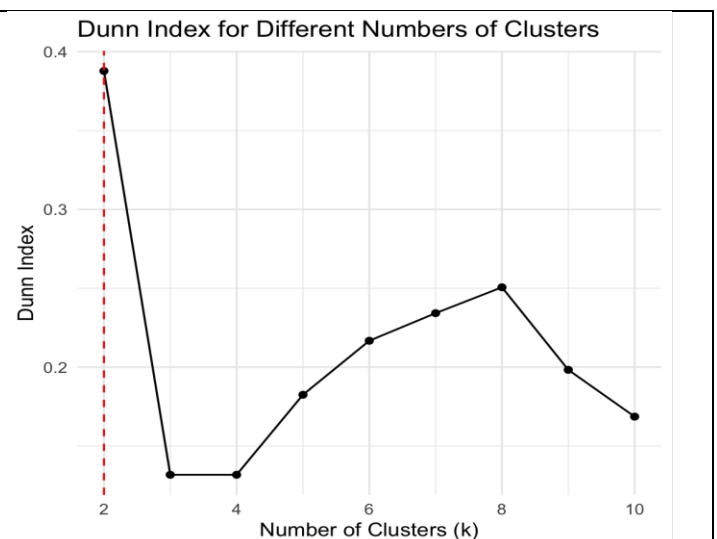


Fig4.Dunn Index

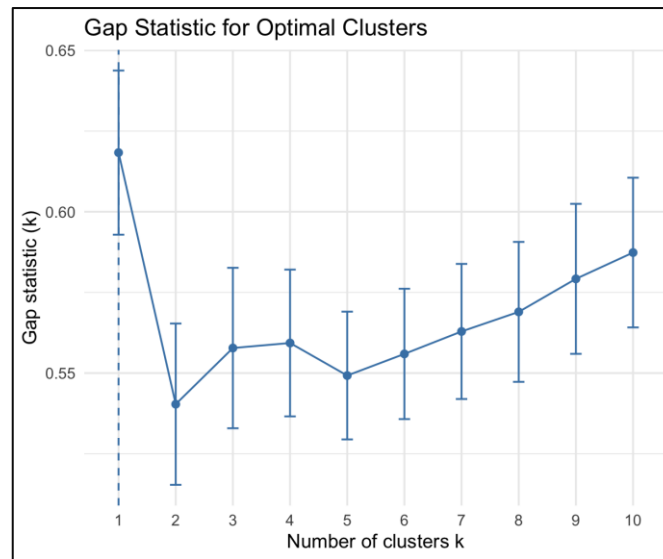


Fig 4. Gap Statistic

To make a firm decision on the number of clusters, I plotted the Gap statistic plot. According to the Gap statistic plot, the optimal number of clusters can be found at the "elbow", namely where the gap statistic first begins to flatten or decrease as the number of clusters increases. Often a choice is made for the smallest  $k$  such that the gap statistic for  $k$  is the maximum before it stabilizes or exhibits marginal further incremental improvements.

In the plot the gap statistics peak at  $k=1$  but since we generally want more than one cluster, we would have to look further. For  $k=2$  through  $k=4$ , the gap statistic is relatively flat with overlapping error bars. At around  $k=5$  or  $k=6$  this value starts increasing more consistently and these could also be plausible choices if the incremental gains are meaningful. Given this, reasonable choices would be  $k=2$  or  $k=5$  based on the overall stability and changes in the gap statistic. To achieve a finer level of granularity in my clustering analysis, I have selected  $K=5$  as the number of clusters.

### Hierarchical-

In clustering, we used the Elbow Method, Silhouette Width, and Dunn Index to determine the best number of clusters. Different insights into the clustering structure were provided by each method. A discernible bend in the WSS (Within-Cluster Sum of Squares) graph, signifying a balance between compactness and separation, led the Elbow Method to propose a possible ideal  $k=3$ . However, as it maximized the average silhouette width and suggested clearly distinct clusters with little overlap, the Silhouette Width



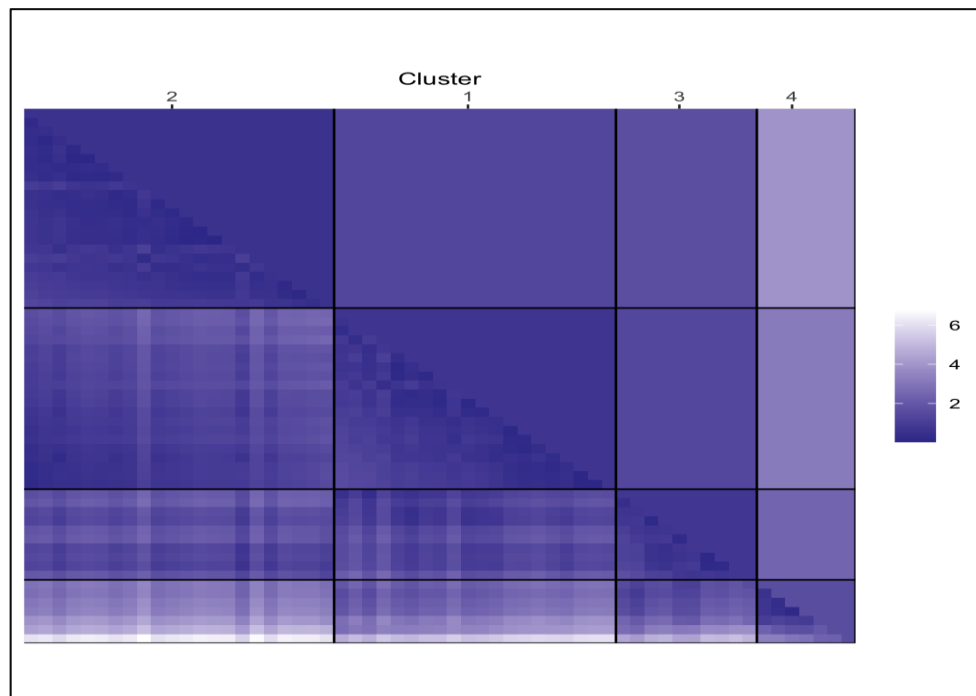
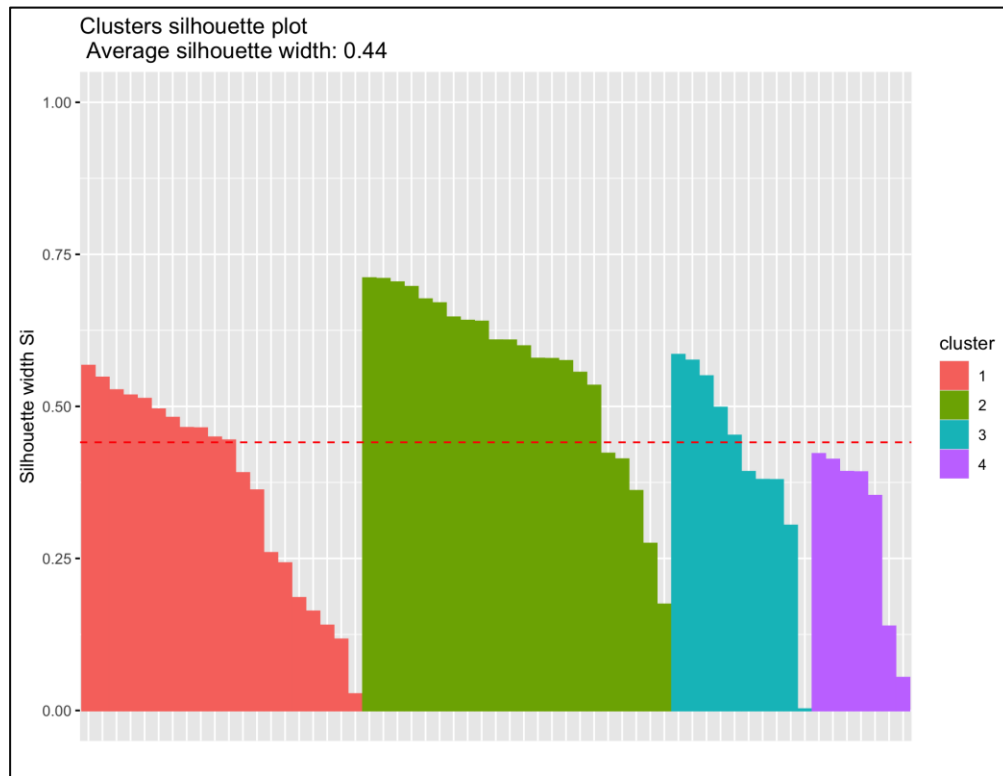
suggested that  $k = 2$  would offer the clearest separation. Conversely, the Dunn Index peaked at  $k=4$ , indicating that four clusters would better distinguish between demographic and economic components by capturing more granularity in the data. Following a review of these findings, we decided to use  $k = 4$  for the thorough feature set analysis. The hierarchical dendrogram, which showed a significant structure with four separate clusters, served as the basis for this choice. A smaller number of clusters might have obscured the specific differences between groups with high Hispanic and poverty levels, high Black population and unemployment rates, high median income, and a variety of demographic profiles, but at  $k = 4$ , we were able to capture these differences. After integrating insights into separation (Dunn Index), distinctiveness (Silhouette), and compactness (Elbow), we determined that  $k = 4$  best satisfies our objectives for this analysis, finding a compromise between statistical measures and the interpretability provided by the dendrogram.

## **Cluster Evaluation, Unsupervised method**

In this part of the report evaluate each cluster by internal validation metrics to assess and compare the clustering results. Each cluster is visualized with a dissimilarity plot, where darker colors indicate greater similarity among data points within the same cluster. In addition, an average silhouette plot is included, showing the average silhouette width for each cluster; a higher silhouette width signifies stronger cohesion within clusters, meaning the points in a cluster are more like each other than to points in other clusters. These metrics help confirm the effectiveness of the clustering by demonstrating distinct groupings with high internal similarity. All the internal validation of clusters is done on clustering results of K-Means clustering.

### **A. Cluster One: Demographic and COVID-19 Impact**

For Cluster one, using four clusters resulted in an average silhouette width of 0.44, indicating moderate cohesion within clusters, as shown in Figure 1.2. Notably, Clusters 1 and 2 contain around 60-65% of New York counties, while Clusters 3 and 4 have fewer counties. Importantly, no cluster has negative silhouette widths, suggesting there was no misclassification of counties, which supports the reliability of the clustering. Based on these observations, we conclude that the chosen clustering method and the number of clusters in the k-means algorithm are appropriate, reinforcing confidence in the assumptions drawn from these clusters.



## B. Cluster Two: Economic Vulnerability

For Cluster 2, using a 4-cluster configuration resulted in an average silhouette width of 0.26, indicating moderate separation between clusters. The dissimilarity plot reveals significant within-cluster similarity, as reflected by darker colors, underscoring the cohesiveness of clusters overall. In examining Cluster 3, we noted a few instances where certain counties exhibited a negative silhouette width. However, with only one county showing signs of potential misclassification, this has a minimal effect on the overall conclusions drawn from the clustering. Consequently, while the chosen number of clusters appears to provide a satisfactory structure, Cluster 1 demonstrated stronger clustering performance. Nonetheless, the insights derived from these clusters remain reliable, offering a solid foundation for further analysis and recommendations.

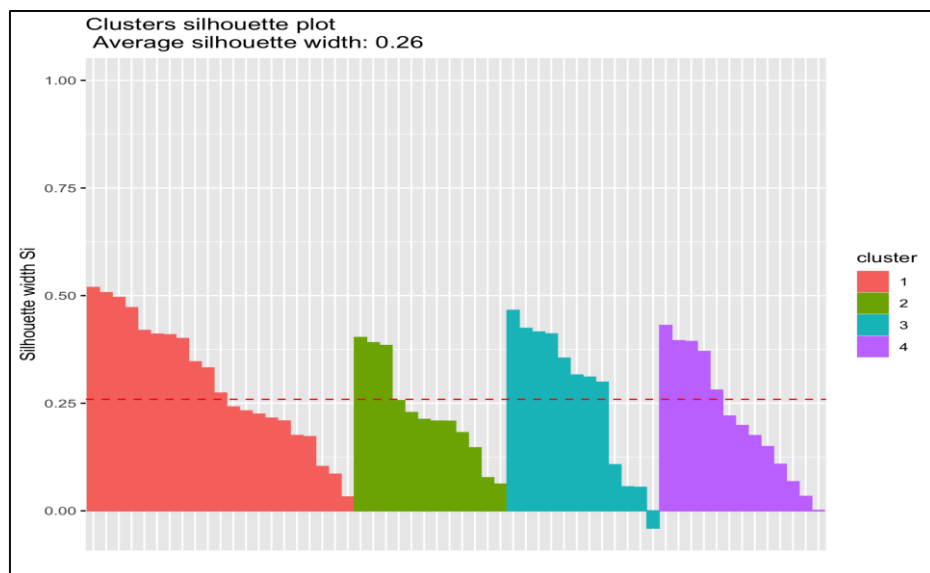
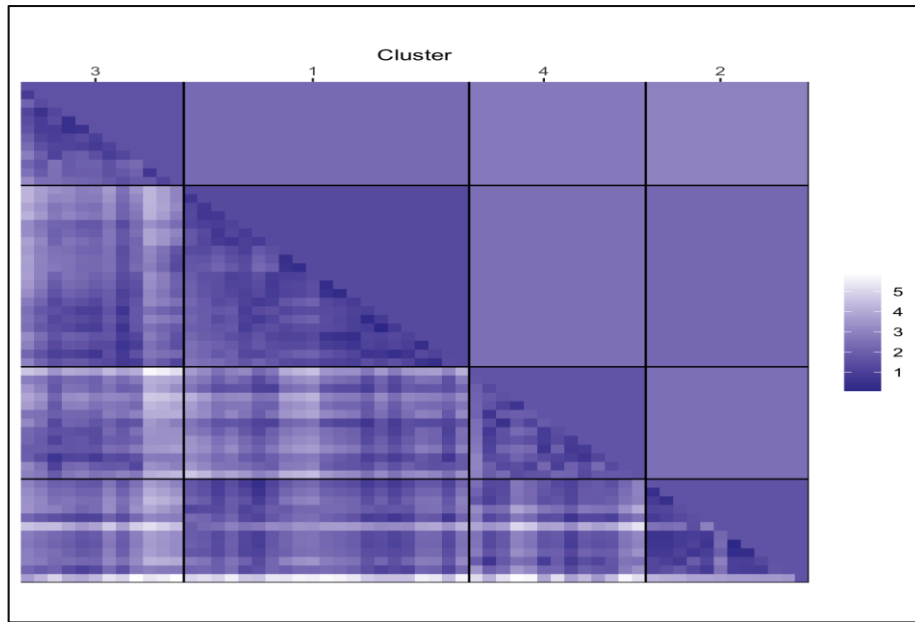
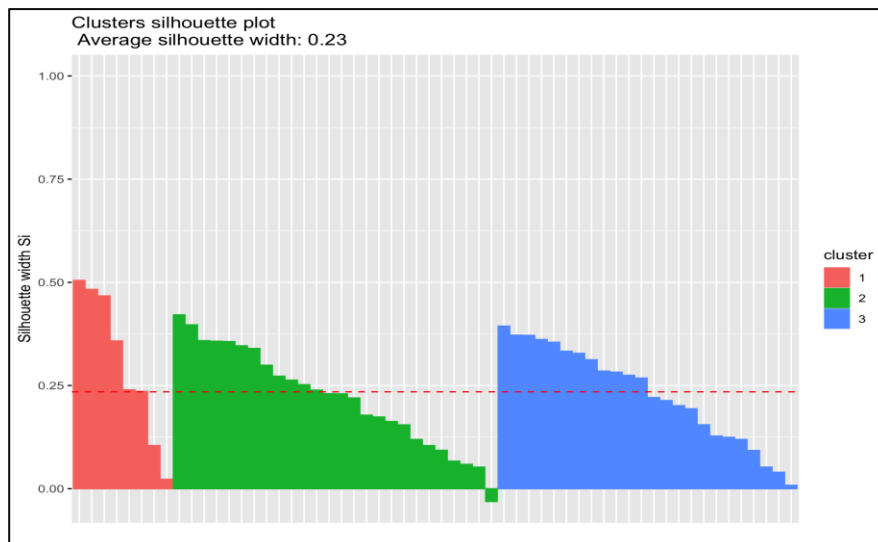


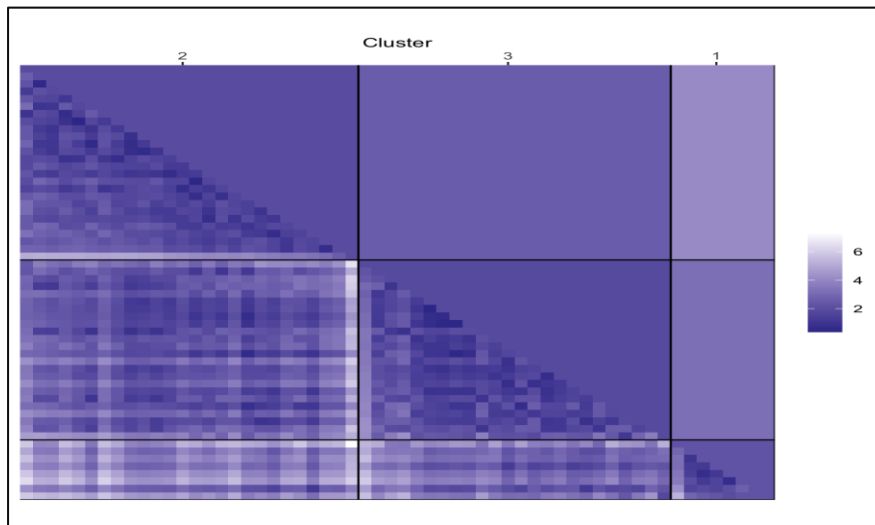
Fig 2.1



### C. Cluster Three: Socioeconomic Hardship and COVID-19 Outcomes

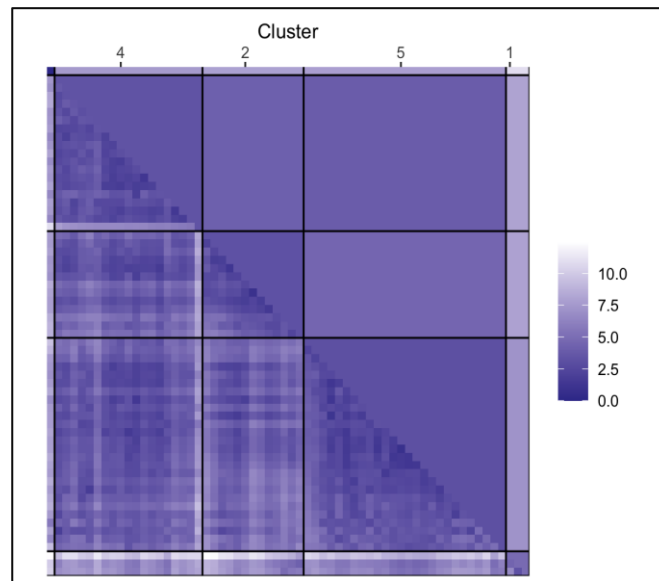
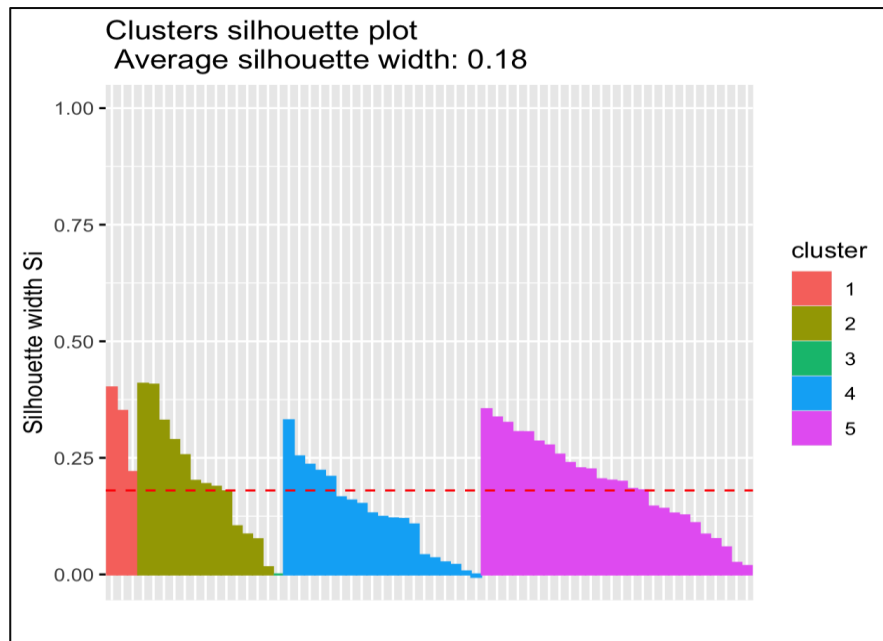
For this cluster using 3 clusters resulted in an average silhouette width of 0.23. In dissimilarity plot darker colors signify higher similarity within clusters. Clusters 2 and 3 show particularly strong silhouette widths. Only one instance of a negative silhouette width appeared in Cluster 2, while the lower silhouette width in Cluster 1 may be due to its few counties, suggesting potential outliers. Notably, in Clustering Plot 3.1, Cluster 1 includes Richmond and Nassau counties, which have exceptionally high death rates, indicating their outlier behavior. Overall, this clustering approach is effective and supports the reliability of the assumptions made. Thus, the chosen number of clusters for the K-means algorithm appears appropriate.





## Cluster Four: Comprehensive Demographic and Economic Clustering

The unsupervised cluster evaluation plots provide insight into the effectiveness and cohesion of the clusters generated. The silhouette plot, with an average silhouette width of 0.18, reveals relatively low cluster cohesion overall, as indicated by the modest height of the bars in most clusters. Higher silhouette values imply better-defined clusters, so this lower value suggests room for improvement in separation. However, clusters 4 and 5 show moderate cohesion, with some segments exceeding 0.25 in silhouette width, indicating reasonable internal similarity. Additionally, the dissimilarity plot illustrates the degree of dissimilarity within clusters, represented by darker colors. The concentration of darker shades along the diagonal highlights a stronger internal similarity within some clusters, notably clusters 4 and 5. These visualizations suggest that while there is internal consistency in some clusters, others may benefit from adjustments in cluster count or feature selection to enhance separation and reduce overlap.

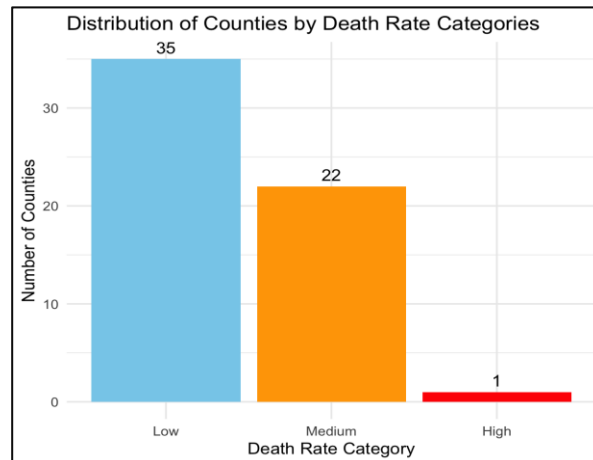


## Supervised Evaluation of Clustering

### Gound Truth Selection-

In this analysis, we take the death percentage of each county to be a ground truth variable in assessing the quality of our clustering models. We chose the ground truth variable-death percentage-because it reflects the severity of the impact from COVID-19 in each county, which is an extremely important concern for our stakeholders-U.S. Government Officials. This is where stakeholders are focusing their efforts in terms of controlling and reducing death rates, hence it should be a relevant and impactful

metric for the clustering analysis. Based on various socio-economic and demographic factors that relate to death rates, insights could be provided that can be useful for the policymaker in proposing specific targeted policy interventions or resource allocations.



To categorize the death percentage for each county, we transformed the continuous variable into discrete labels by dividing it into three categories: Low, Medium, and High death rates (As Shown in the above plot). This was accomplished using the cut function, which segmented deaths\_percent based on predefined ranges, thus creating a variable (death label) that represents our ground truth.

Then, we used the Rand Index, which serves to evaluate our clustering assignments against the ground truth labels. The Rand Index, implemented through the adjustedRandIndex function in the mclust package, compares pairs of observations to determine whether they are clustered similarly in both the model and the ground truth. For the 4 Clusterings we are getting the Rand Index as follows.

| K-Means Clustering | Rand Index |
|--------------------|------------|
| Clustering 1       | 0.62       |
| Clustering 2       | 0.35       |
| Clustering 3       | 0.15       |
| Clustering 4       | 0.17       |

The Rand Index values of the four different clustering depicts that the Clustering 1-with the addition of Hispanic and Black population percentages with death percentage-performs best with a Rand Index of

0.62. In this clustering approach, the variation seems to express moderate agreement with the existing death rate categories, which further indicates that demographic factors related to race/ethnicity can result in more significant variations in death percentages than socio-economic indicators alone.

The Rand Index of 0.35 for this clustering is still viewed as a satisfactory alignment with the death rate categories. Hence, Poverty and income inequality, represented through the Gini index, do reflect meaningful patterns related to death rates. Such a result would indicate that socio-economic factors do bear relevance to understanding death rates. However, the Clustering 3 and Clustering 4 approaches, with Rand Index values of 0.15 and 0.17 respectively, show weaker alignment, suggesting that economic indicators alone or a broad mix of demographic and economic indicators may not be as effective for clustering by death rates. This might also suggest that we need to apply different types of clustering algorithms on the clusters where the rand index is currently very low and try to evaluate the clustering again. We have done this in our exceptional work of the project.

### **Exceptional Work:**

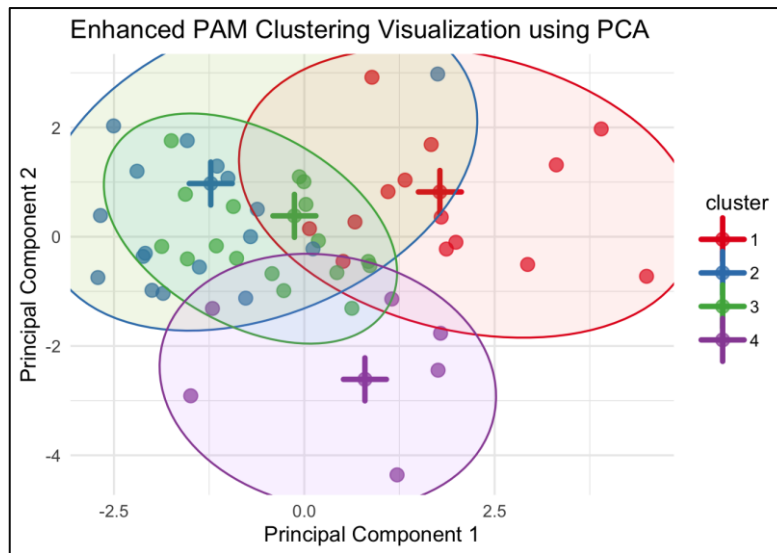
#### **Partitioning Around Medoids (PAM) Clustering:**

Since the Rand index of the Cluster 4 is very low, we will try to explore other Clustering algorithms and try to examine if we can get better clustering and better evaluation results both in supervised and unsupervised learning. Our goal is to create clustering in such a way that it performs better on the supervised and unsupervised clustering evaluation. Hence, we will be using Partitioning Around Medoids (PAM) method to do the clustering.

For PAM Clustering Visualization using Principal Component Analysis (PCA), the plot provides a clear view of how counties are grouped into four distinct clusters based on their socio-economic and demographic factors, including poverty rates, racial demographics, and income levels. The ellipses around each cluster help define the boundaries and overlap of the clusters, while the cross-shaped medoids represent the central points within each cluster, acting as representative data points.

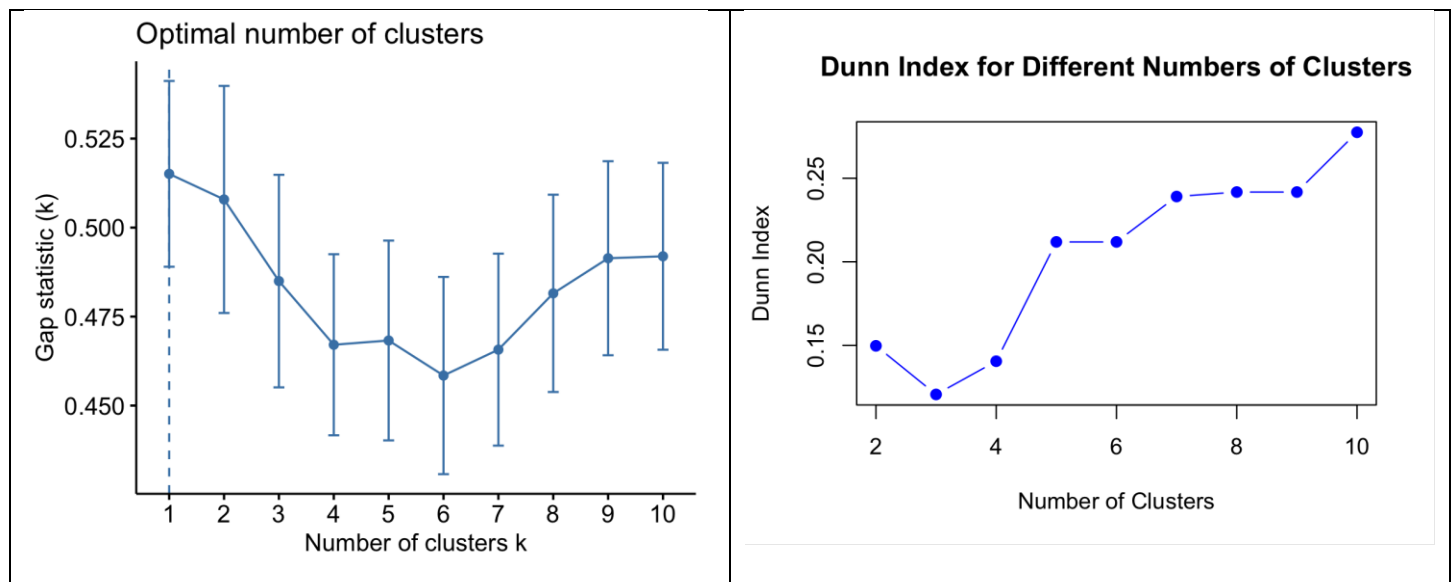
Each cluster shows a specific distribution between the two principal components, which encapsulates the main variance in the dataset. For instance, clusters with more overlap may indicate similarities among counties in certain factors, while distinct clusters suggest unique socio-economic profiles that set these groups apart. This clustering can aid stakeholders in identifying targeted intervention areas based on the socio-economic and racial characteristics associated with different death rates across counties.





## Deciding the number of clusters

To determine the number of clusters for the PAM method, we used both Dunn Index and Gap Statistic approach.

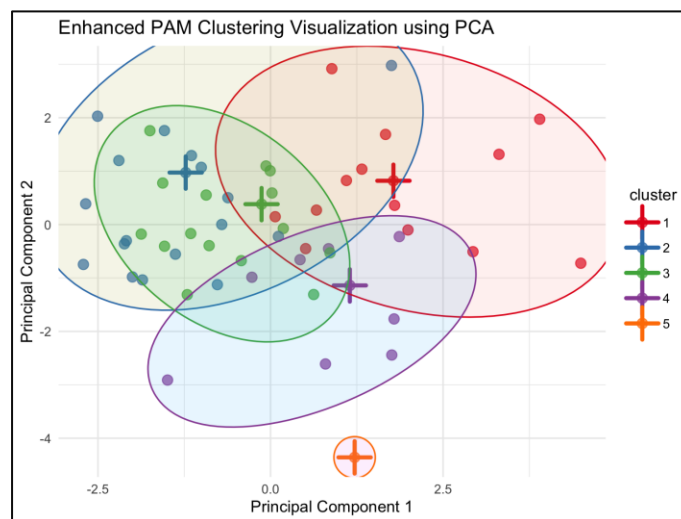


Higher Dunn Index values generally indicate better-defined clusters, and this trend suggests that increasing the clusters improves the clustering quality, with 5 clusters being a good balance point before diminishing returns.

The Gap Statistic plot shows a general decrease in the gap value as the number of clusters increases, but

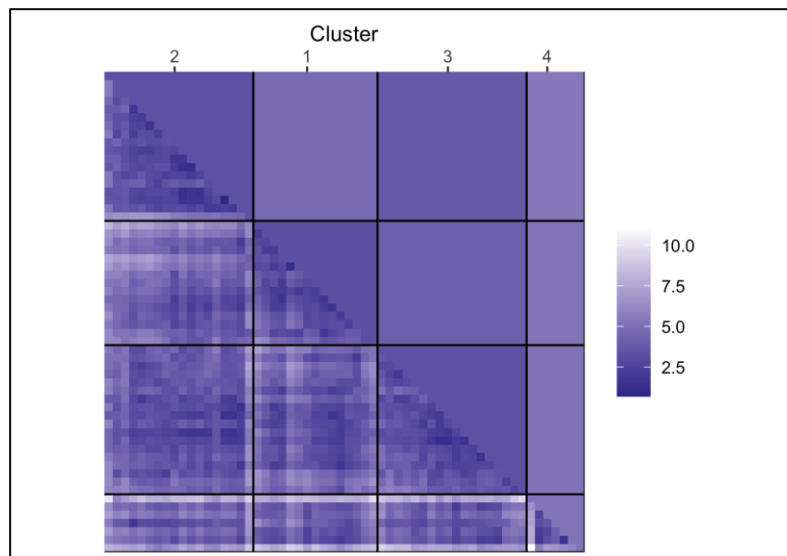
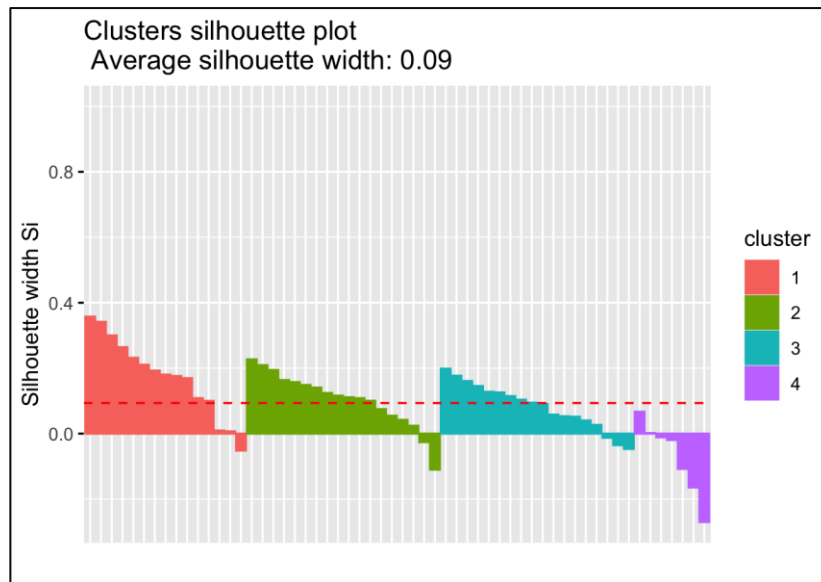
with noticeable variation in error bars. The plot shows the highest gap statistic around 2 or 3 clusters, suggesting that fewer clusters could yield distinct group separation. However, given the large variability, 5 clusters also remain a reasonable choice based on a trade-off between separation and within-cluster homogeneity.

On Choosing 5 clusters we were getting below cluster plot, with 5<sup>th</sup> cluster having only one county. This clustering has been done after removing the outliers like Bronx County, Queens County, New York County and Kings County (since these counties are massive in terms of population density and it is suggested to the stakeholders to look at these counties separately), Hence we don't want to remove any other counties. Therefore, we decided to cluster with 4 clusters to achieve a more meaningful segmentation of the data, balancing interpretability and compactness.



## Unsupervised Cluster Evaluation

In the case of employing 4 clusters, the average silhouette width was determined to be 0.09, as depicted in below figure. Notably, the similarity within clusters was not very good. Examining this clustering approach, it is observed that the cluster exhibits 4 instances of negative silhouette width. The overall silhouette width in these clusters appears average. Consequently, we have confidence in the adequacy of the chosen number of clusters for PAM and in the validity of the assumptions derived from these clusters.



## Supervised Cluster evaluation

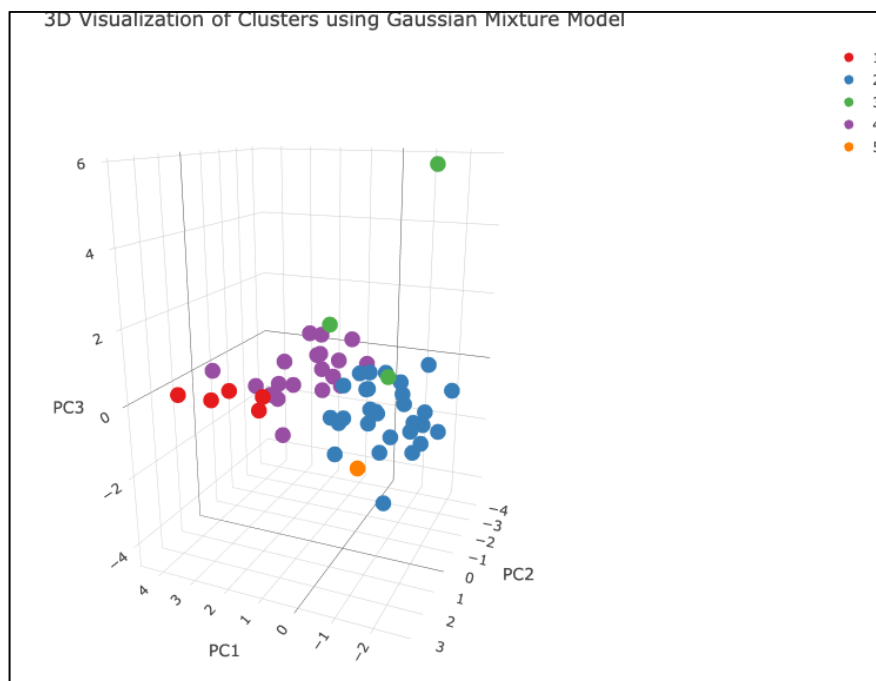
In this analysis, we again take the death percentage of each county to be a ground truth variable in assessing the quality of our clustering models. We chose the ground truth variable-death percentage- because it reflects the severity of the impact from COVID-19 in each county, which is an extremely important concern for our stakeholders-U.S. Government Officials.

The Adjusted Rand Index (ARI) for PAM clustering relative to the death rate categories was calculated as 0.063. This low ARI score suggests limited alignment between the clustering solution and the death rate categories (Low, Medium, High). Therefore, the clusters identified by the PAM algorithm may reflect

patterns that are not strongly related to the death rate, potentially indicating that the chosen features are capturing other aspects of the data distribution rather than directly correlating with death rate categories.

### **Gaussian mixture models Clustering:**

Since we received more negative results using PAM, we will move on to the Gaussian Mixture Model Clustering. One of the key features of GMM is that it can determine the optimal number of clusters automatically. By leveraging a metric called the Bayesian Information Criterion (BIC), GMM can evaluate multiple models with different cluster counts and select the one with the best fit. This eliminates the need to use external methods like Within-Cluster Sum of Squares (WCSS) or the Dunn Index for determining the optimal number of clusters, simplifying the clustering process while ensuring robust results. This feature makes GMM a flexible and efficient choice for data segmentation, especially when the number of clusters is unknown.



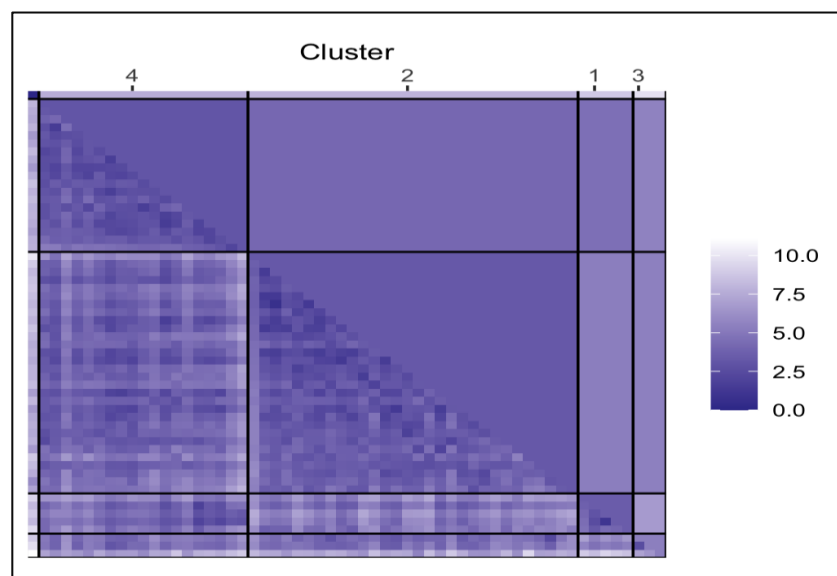
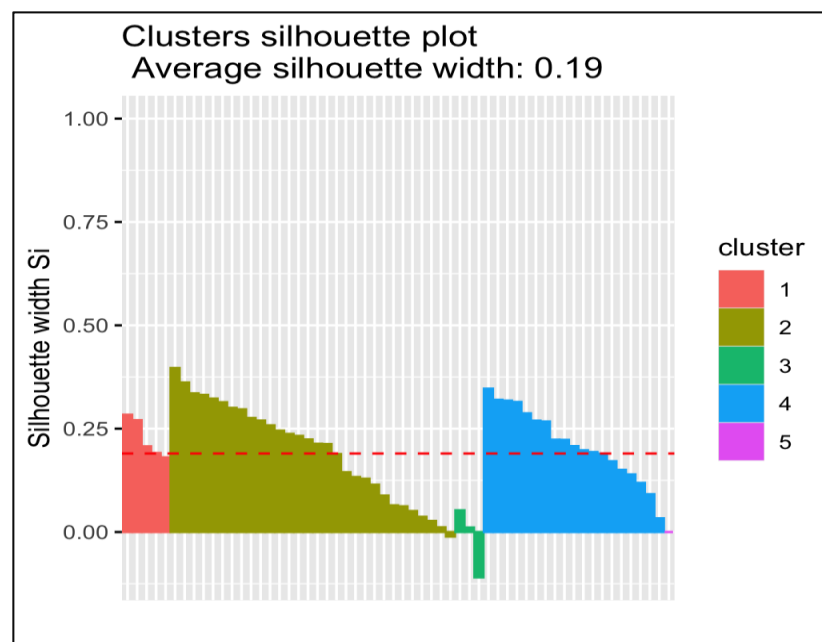
Looking at the plot, we can infer that it is giving similar clustering results as PAM. We can clearly observe that the GMM clustering decided to cluster the counties with 5 clusters. In our report under PAM section, we derived the number of clusters to 5 with Dunn Index and Gap statistics, thus we decided to skip the part where we find the number of clusters.

### **Unsupervised Evaluation-**

In the analysis using Gaussian Mixture Model clustering, the silhouette plot and dissimilarity matrix

provide insights into cluster cohesiveness and separation. The silhouette plot shows an average silhouette width of 0.19, indicating moderate cohesion within clusters. Most clusters have positive silhouette widths, showing effective clustering, while clusters with lower widths or negative values, like in the green cluster, suggest overlap or lower separation quality.

The dissimilarity matrix plot reveals darker regions within cluster boundaries, showing strong intra-cluster similarities, especially in clusters with higher silhouette scores. The relatively well-separated blocks in the matrix illustrate the GMM's ability to form distinct groups, supporting the reliability of the chosen clustering structure without requiring external metrics for cluster validation, like WCSS or the Dunn index.



## Supervised Evaluation

Since we got a better silhouette width of 0.19, we expected that the Rand Index would improve. But contrary to our predictions, the Rand Index for GMM clustering is 0.031, which is less than the PMM clustering Rand Index. This suggests that the clusters formed by the GMM do not strongly correlate with the death rate categories. The low ARI score could imply that the features used for clustering may not sufficiently capture the distinctions in death rates, or that the GMM approach may not be optimal for this data in terms of aligning with the specific death rate labels.

## Conclusion and Recommendations

In this report, we conducted an in-depth analysis of county-based census data in New York State to investigate the intricate relationships between demographics, socioeconomic factors, and the spread and containment of COVID-19, with the help of K-means Clustering. Moreover, we expanded our analysis to the selected counties of different states like New Jersey, California, and Oregon from our project report 1, with the help of Hierarchical clustering. After the evaluation of the K Means clustering, we concluded that our First Cluster has the best set of features having the highest Rand Index and Average Silhouette Width Number. Cluster Four and Three on the other hand have resulted us to conclude that as we add more features for clustering the average Silhouette Width and Rand Index reduces significantly. In our quest to find better clustering we expanded our analysis by performing PAM and GMM clustering on Cluster Four.

Based on this analysis, we suggest to our stakeholder, which are USA Government Officials and New York County Officials that the counties with higher Hispanic and Black population percentages faced disproportionate death rates, particularly in denser, urban areas with socioeconomic challenges. To address these disparities in future pandemics or health crises, the following steps are recommended:

**Targeted Healthcare Access Improvements:** Enhance healthcare infrastructure in densely populated urban counties, particularly those with higher minority populations.

**Support for Essential Workers:** Implement policies to protect essential workers, who are often from minority communities and who are earning less than 15000\$, to reduce occupational exposure. Also provide critical workers in affected demographic sectors with occupational health measures, such as

personal protective equipment (PPE) and vaccines.

**Public Health Outreach and Education:** Increase community outreach and education in counties with high Hispanic and Black populations to ensure these communities are well-informed about healthcare resources and preventive measures.

**Resource Allocation for Rural Areas:** Counties with high death rates but low minority populations, like those in Cluster 1, may benefit from targeted healthcare resources for vulnerable populations, such as elderly residents.

**Enhance Access for Elderly and Rural Communities:** In rural counties with larger senior populations and less access to healthcare, make use of telemedicine, mobile health units, and regional hospital support.

**Tailored Emergency Responses for High-Risk Demographics:** Create culturally appropriate outreach to increase access to healthcare in counties with large Black and Hispanic populations. This can be done through local partnerships and bilingual resources.

**Tailored Support for Outliers:** The counties such as New York county, Queens, Kings and Richmond County, would definitely require a sperate attention and specialized teams should be created to tackle the challenges in these counties which has not only high Black and Hispanic population density, but they also have high poverty rate and high-Income disparity which has ultimately lead to high death percentages.

This analysis suggests to our stakeholder, which are USA Government Officials and New York County Officials that while population density and healthcare access are critical factors in managing COVID-19 outcomes, addressing socioeconomic and racial disparities can significantly reduce mortality in future health emergencies.

## Appendix

[https://mhahsler.github.io/Introduction\\_to\\_Data\\_Mining\\_R\\_Examples/book/cluster-analysis.html#gaussian-mixture-models](https://mhahsler.github.io/Introduction_to_Data_Mining_R_Examples/book/cluster-analysis.html#gaussian-mixture-models)

### Exceptional Code -

## Supervised Evaluation

```
cases_NY_select_without_outlier$death_label <- cut(cases_NY_select_without_outlier$deaths_percent,
  breaks = 3, # Adjust the number of bins as appropriate
  labels = c("Low", "Medium", "High"))
```

```
# Plot the distribution of death rate categories
ggplot(cases_NY_select_without_outlier, aes(x = death_label, fill = death_label)) +
  geom_bar(stat = "count") + # Use stat = "count" to count the number of counties in each category
  geom_text(stat = 'count', aes(label = ..count..), vjust = -0.5) + # Display the count above each bar
  labs(title = "Distribution of Counties by Death Rate Categories",
        x = "Death Rate Category",
        y = "Number of Counties") +
  scale_fill_manual(values = c("Low" = "skyblue", "Medium" = "orange", "High" = "red")) +
  theme_minimal() +
  theme(legend.position = "none")
```

```
# Rand Index calculation for each clustering result
# Ensure `death_label` is a factor for comparison
ground_truth <- as.numeric(as.factor(cases_NY_select_without_outlier$death_label))
```

```
# Rand Index for the first clustering result (you can repeat this for others)
rand_index_2 <- adjustedRandIndex(ground_truth, km_clusters)
print(paste("Rand Index for first clustering:", rand_index_1))
```

```
# Output Rand Index values to compare each clustering result
print(paste("Rand Index for clustering 2:", rand_index_2))
```

### **For 3D Plotting -**

```
# Fit the Gaussian Mixture Model
m <- Mclust(scaled_features)

# Perform PCA for dimensionality reduction
pca_result <- prcomp(scaled_features, scale. = TRUE)
pca_data <- as.data.frame(pca_result$x[, 1:2]) # Take the first 2 principal components
pca_data$cluster <- as.factor(m$classification) # Add the cluster assignments
```



```
library(plotly)

# Take the first 3 principal components
pca_data_3d <- as.data.frame(pca_result$x[, 1:3])
pca_data_3d$cluster <- as.factor(m$classification)

# 3D plot
plot_ly(pca_data_3d, x = ~PC1, y = ~PC2, z = ~PC3, color = ~cluster, colors = 'Set1') %>%
  add_markers() %>%
  layout(scene = list(xaxis = list(title = 'PC1'),
    yaxis = list(title = 'PC2'),
    zaxis = list(title = 'PC3')),
    title = "3D Visualization of Clusters using Gaussian Mixture Model")
```